MIL-M-28787/224A(EC)
29 June 1982
SUPERSEDING
MIL-M-28787/224(EC)
14 July 1976

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE JDF

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic digital JK flip-flop module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.187 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JDF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/224-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

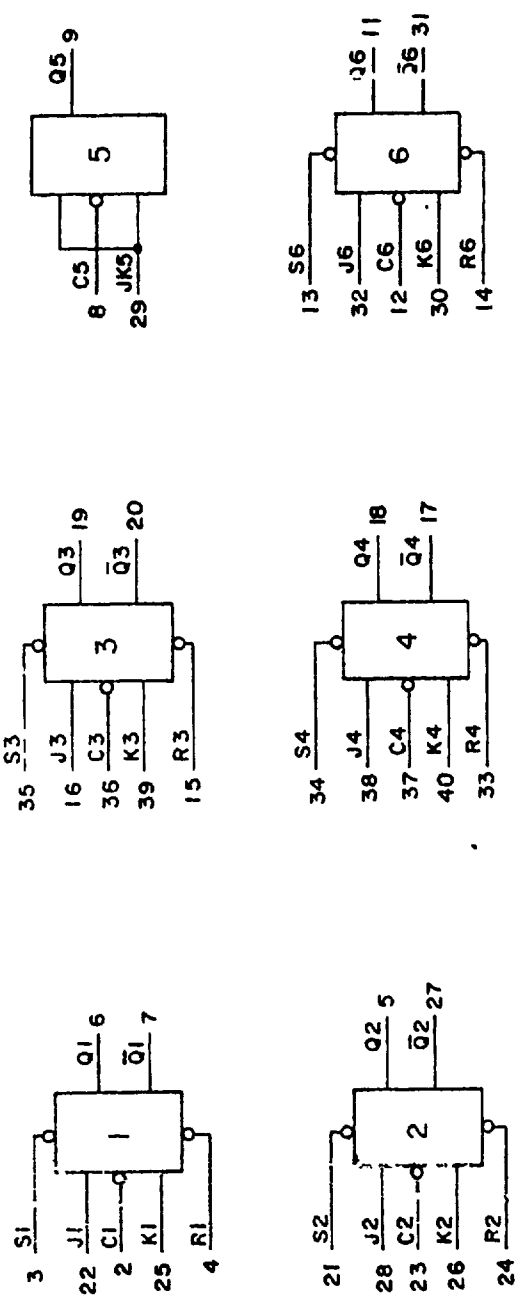
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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NOTES:
1. J, K = data inputs
2. C = clock input
3. S = set input
4. R = reset input
5. Q = data output
6. $\overline{Q}$ = complement of Q data output.

FIGURE 1. JDF module function diagram, JK flip-flop.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B11	V_{OUT} (low - loaded)	IV	S	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	V
C1-C26	Input breakdown current	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C27	Input breakdown current (JK5)	V	S	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	mA
D1-D11	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1 E27	Input clamp diode voltage	VII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
F1-F10	Functional	VIII	X	Test vectors must produce outputs as specified in test table.								
G1-G11	Static output voltage, low level, and input threshold check	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H11	Static output voltage, high level, and input threshold check	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J10	Input current (high level), J and K	X	X	0	50	0	50	0	50	0	50	µA
J11-J20	Input current (high level), S and R	X	X	0	100	0	100	0	100	0	100	µA
J21	Input current (high level), JK5	X	X	0	100	0	100	0	100	0	100	µA
J22-J27	Input current (high level), C	X	X	-1.0	0.2	-1.0	0.2	-1.0	0.2	-1.0	0.2	mA
K1-K10	Input current (low level), J and K	XI	X	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	mA
K11-K20	Input current (low level), S and R	XI	X	-0.2	-4.4	-0.2	-4.4	-0.2	-4.4	-0.2	-4.4	mA
K21	Input current (low level), JK5	XI	X	-0.2	-4.4	-0.2	-4.4	-0.2	-4.4	-0.2	-4.4	mA
K22-K27	Input current (low level), C	XI	X	-0.3	-6.9	-0.3	-6.9	-0.3	-6.9	-0.3	-6.9	mA
L	Power supply current	XII	X	30	228	30	228	30	228	30	228	mA
M1-M11	Transition time (t_{TLH}), C_L = 50 pF	XIII	S	3	20	3	20	3	25	3	25	ns
M12-M22	Transition time (t_{TLH}), C_L = 150 pF	XIII	Q	5	30	5	30	5	35	5	35	ns
N1-N11	Transition time (t_{THL}), C_L = 50 pF	XIII	S	3	20	3	20	3	25	3	25	ns
N12-N22	Transition time (t_{THL}), C_L = 150 pF	XIII	Q	5	25	5	25	5	30	5	30	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P1-P11	Propagation delay time (t_{PLH}), clock, C_L = 50 pF	XIII	S	5	18	5	18	5	20	5	20	ns
P12-P22	Propagation delay time (t_{PLH}), clock, C_L = 150 pF	XIII	Q	5	28	5	28	5	30	5	30	ns
Q1-Q11	Propagation delay time (t_{PHL}), clock, C_L = 50 pF	XIII	S	5	27	5	27	5	29	5	29	ns
Q12-Q22	Propagation delay time (t_{PHL}), clock, C_L = 150 pF	XIII	Q	5	37	5	37	5	39	5	39	ns
R1-R10	Propagation delay time (t_{PLH}), set and reset, C = low, C_L = 50 pF	XIV	S	5	15	5	15	5	17	5	17	ns
R11-R20	Propagation delay time (t_{PLH}), set and reset, C = low, C_L = 150 pF	XIV	Q	5	25	5	25	5	27	5	27	ns
S1-S10	Propagation delay time (t_{PHL}), set and reset, C = low, C_L = 50 pF	XIV	S	5	38	5	38	5	40	5	40	ns
S11-S20	Propagation delay time (t_{PHL}), set and reset, C = low, C_L = 150 pF	XIV	Q	5	48	5	48	5	50	5	50	ns
T1-T10	Propagation delay time (t_{PHL}), set and reset, C = high, C_L = 50 pF	XIV	S	5	28	5	28	5	30	5	30	ns
T11-T20	Propagation delay time (t_{PHL}), set and reset, C = high, C_L = 150 pF	XIV	Q	5	38	5	38	5	40	5	40	ns
U1-U6	Maximum operating frequency and minimum clock pulse width check (PW measurement)	XV	S	27	50	27	50	27	50	27	50	ns
U7-U11	Maximum operating frequency and minimum clock pulse width check (PRP measurement)	XV	S	54	66	54	66	54	66	54	66	ns
V1-V6	Input low-level setup and hold time check (V_{OL} measurement)	XVI	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
W1-W6	Maximum transition time check (PW measurement)	XVII	S	960	1,040	960	1,040	960	1,040	960	1,040	ns
X1-X10	Output crosstalk, low level	XVIII	Q	0	680	0	680	0	680	0	680	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
X11-X14	Output crosstalk, low level	XVIII	Q	0	360	0	360	0	360	0	360	mV
X15-X18	Output crosstalk, low level	XVIII	Q	0	320	0	320	0	320	0	320	mV
X19,X20	Output crosstalk, low level	XVIII	Q	0	310	0	310	0	310	0	310	mV
Y1-Y10	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
Z	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV
AA	Power supply filter capacitance	XXI	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
BB1-BB26	Input capacitance (J, K, R, S, C)	XXII	Q	2	15	2	15	2	15	2	15	pF
BB27	Input capacitance (JK5)	XXII	Q	4	30	4	30	4	30	4	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever, is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through BB in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through AA shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through BB shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/224 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833698. Part number M28787/224-1 supersedes part numbers 2833698 and M28787/224.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

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Maximum power dissipation	1.25 W (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current (J and K, except JK5)	
Low level (V_{IL} = 0.0 V)	-2.2 mA maximum
(V_{IL} = 0.4 V)	-2.0 mA maximum
High level (V_{IH} = 2.4 V)	50 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum
Input current (JK5)	
Low level (V_{IL} = 0.0 V)	-4.4 mA maximum
(V_{IL} = 0.4 V)	-4 mA maximum
High level (V_{IH} = 2.4 V)	100 µA maximum
(V_{IH} = 5.5 V)	2 mA maximum
Input current (S and R)	
Low level (V_{IL} = 0.0 V)	-4.4 mA maximum
(V_{IL} = 0.4 V)	-4 mA maximum
High level (V_{IH} = 2.4 V)	100 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current (C)	
Low level (V_{IL} = 0.0 V)	-6.9 mA maximum
(V_{IL} = 0.4 V)	-6.4 mA maximum
High level (V_{IH} = 2.4 V)	-1.0 mA minimum, 0.2 mA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Output voltage	
Low level (I_{OL} = 20 mA)	0.4 V maximum
High level (I_{OH} = -500 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	20 mA
High level (V_{OH} = 2.4 V)	-500 µA
Output short circuit current	-40 mA minimum, -100 mA maximum
Input clamp diode voltage	
I_{CLAMP} = -8 mA	-1.5 V maximum
Propagation delay time	
Clock to Q or $\overline{Q}$	
t_{PLH} (C_L = 50 pF)	5 ns minimum, 20 ns maximum
(C_L = 150 pF)	5 ns minimum, 30 ns maximum
t_{PHL} (C_L = 50 pF)	5 ns minimum, 29 ns maximum
(C_L = 150 pF)	5 ns minimum, 39 ns maximum
Set or reset to Q or $\overline{Q}$	
t_{PLH} (C_L = 50 pF)	5 ns minimum, 17 ns maximum
(C_L = 150 pF)	5 ns minimum, 27 ns maximum
(clock high)	
t_{PLH} (C_L = 50 pF)	5 ns minimum, 30 ns maximum
(C_L = 150 pF)	5 ns minimum, 40 ns maximum
(clock low)	
t_{PHL} (C_L = 50 pF)	5 ns minimum, 40 ns maximum
(C_L = 150 pF)	5 ns minimum, 50 ns maximum
Maximum operating frequency	
C_L = 50 pF	33 MHz minimum

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Triggering edge	High to low
Pulse width	
Clock high level	12 ns minimum
CLock low level	18 ns minimum
Set and reset	20 ns minimum
J, K setup time	
High level before clock high to low transition	12 ns minimum
Low level before clock high to low transition	15 ns minimum
J, K hold time	
Time after clock high to low transition	0 ns minimum
Clock transition time	150 ns maximum

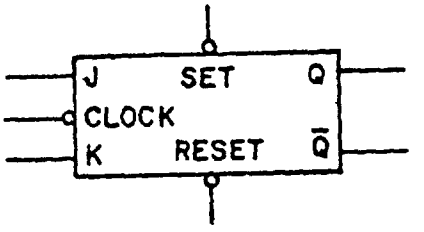
6.5 Module function. This module contains six flip-flops. Five of these are JK flip-flops. Each flip-flop is negative-edge triggered with individual J, K clock and a synchronous set and reset input to each flip-flop. When the clock goes high, the inputs are enabled and data will be accepted. The logical state of the J and K inputs are allowed to change when the clock pulse is in the high state and the stable state will perform according to the truth table as long as minimum set-up times are observed. Input data is transferred to the outputs on the negative edge of the clock pulse. The sixth flip-flop is a toggle type with no reset capability and only the true state is brought out.

6.5.1 Input-to-output relationship. Each JK flip-flop has the input-to-output relationship of table II for positive logic.

TABLE II. Input-to-output relationship for each JK flip-flop.

Inputs					Outputs	
Set	Clear	Clock	J	K	Q	$\overline{Q}$
H 1/	L 2/	X 3/	X	X	L	H
L	H	X	X	X	H	L
L	L	X	X	X	H 6/	H 6/
H	H	↓ 4/	L	L	Q_0 5/	$\overline{Q}_0$ 5/
H	H	↓	H	L	H	L
H	H	↓	L	H	L	H
H	H	↓	H	H	Toggle	

1/ H = high-level voltage (steady state).
2/ L = low-level voltage (steady state).
3/ X = don't care (any input, including transitions).
4/ ↓ = transition from high to low.
5/ Q_0, $\overline{Q}_0$ = the level of Q, $\overline{Q}$ respectively, before the indicated steady-state input conditions were established.
6/ This configuration is unstable; that is, it will not persist when set or clear returns to its inactive (high-level) state. This condition is not tested for nor guaranteed.



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6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Crosstalk relationship. The module has the following module and circuit crosstalk relationships.

6.5.3.1 Module crosstalk. When connected in the worst case mode of operation (all but one output switching simultaneously from a high to a low level) the JDF module could exhibit substantial crosstalk. When the quiet output is a low level, there may be as much as a 680 mV positive-going spike coincident with the negative edge of the switching outputs.

6.5.3.2 Circuit crosstalk. Integrated circuits 2 and 3 can induce as much as 360 mV of noise onto any integrated circuit 1 output. Integrated circuits 1 and 3 induce as much as 320 mV of noise onto any integrated circuit 2 output. Integrated circuits 1 and 2 can induce as much as 310 mV of noise onto any integrated circuit 3 output.

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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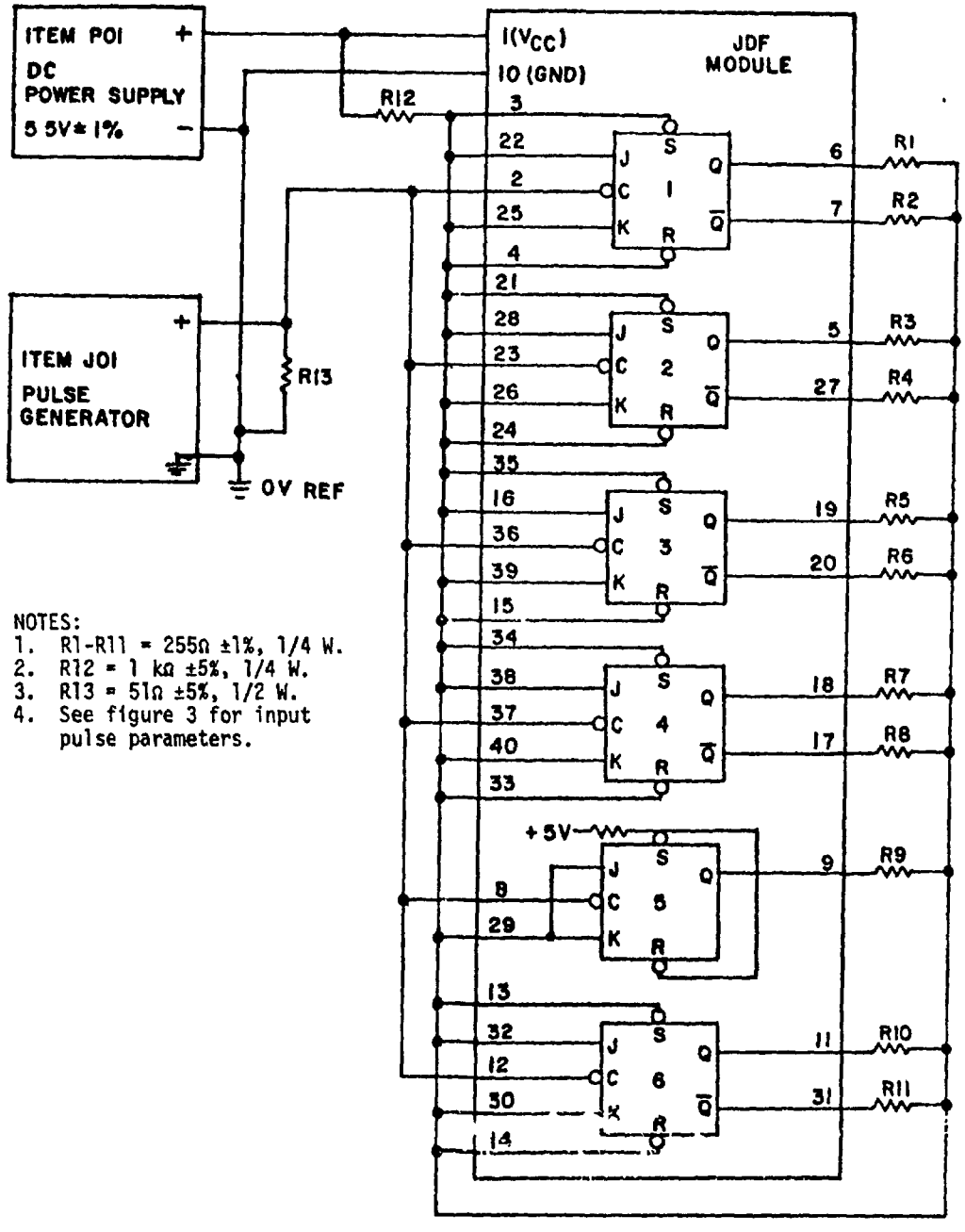
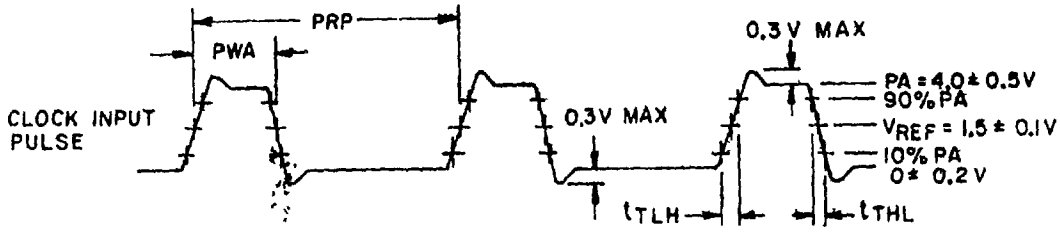


FIGURE 2. Operating temperature and life test circuit.

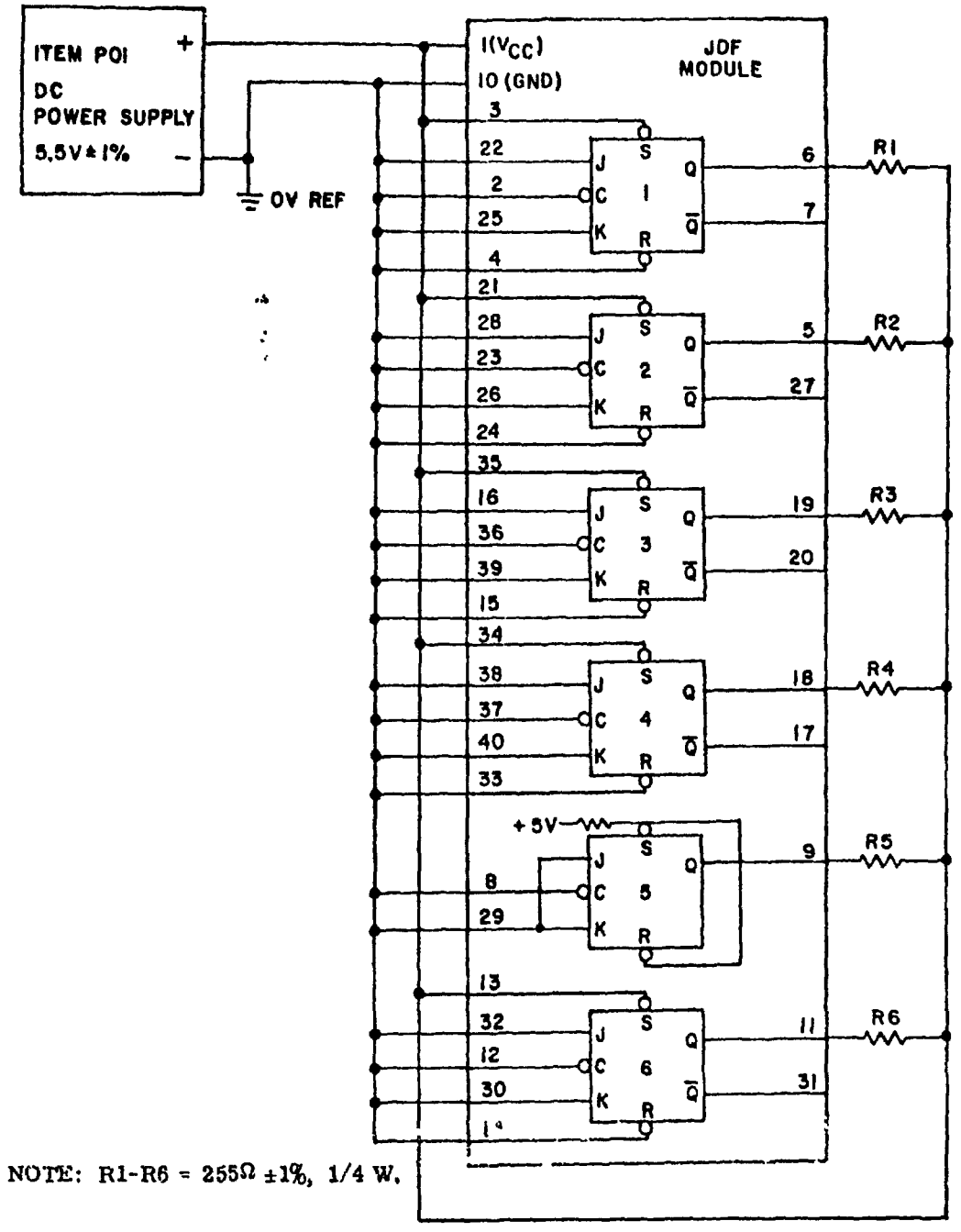
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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 250 ±50 ns.
   $t_{TLH} = t_{THL} = 25 \pm 20$ ns.
2. All voltages are referenced to ground.

FIGURE 3. Operating temperature and life test circuit waveform.

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NOTE: R1-R6 = 255Ω ±1%, 1/4 W.

FIGURE 4. Component temperature test circuit.

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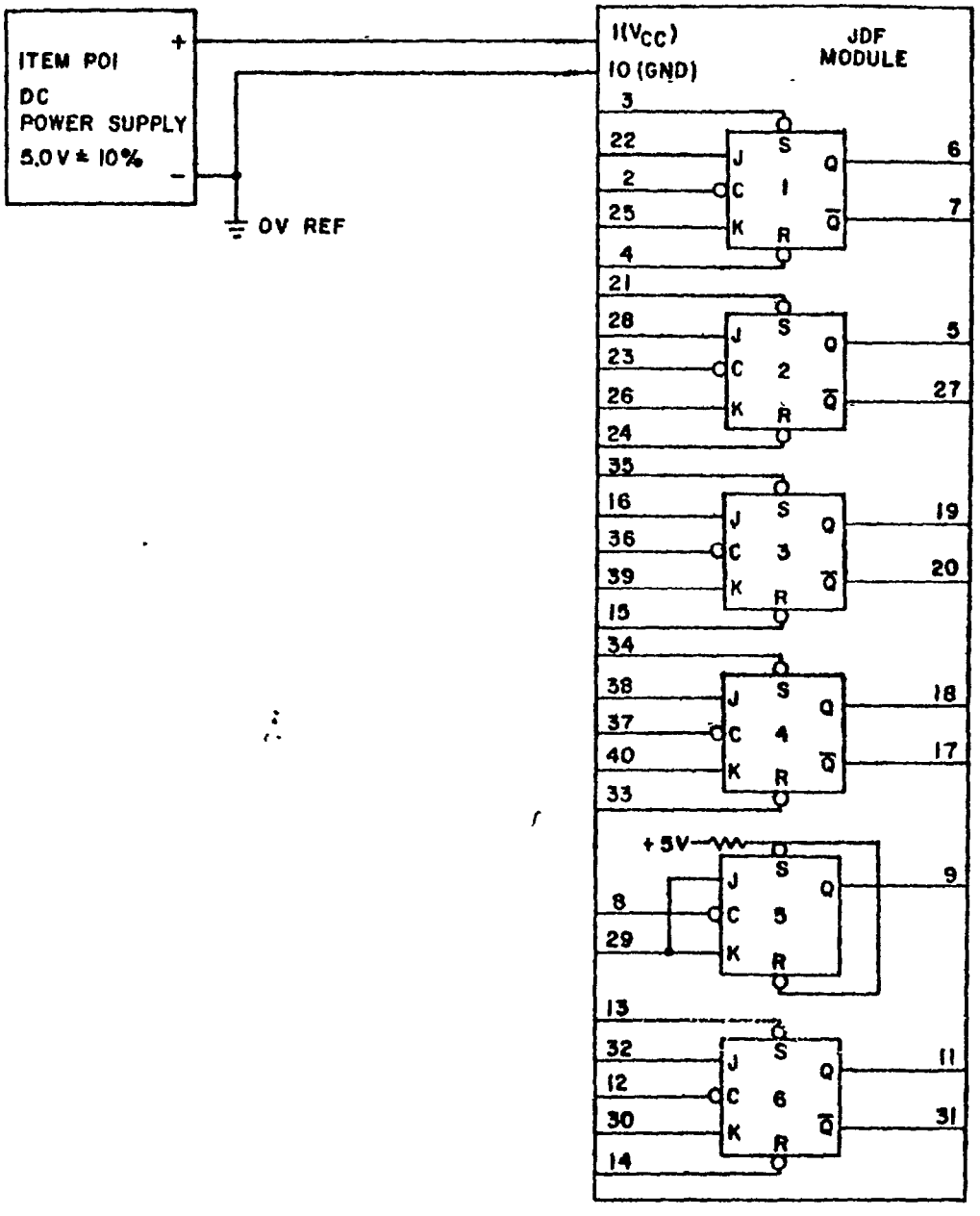


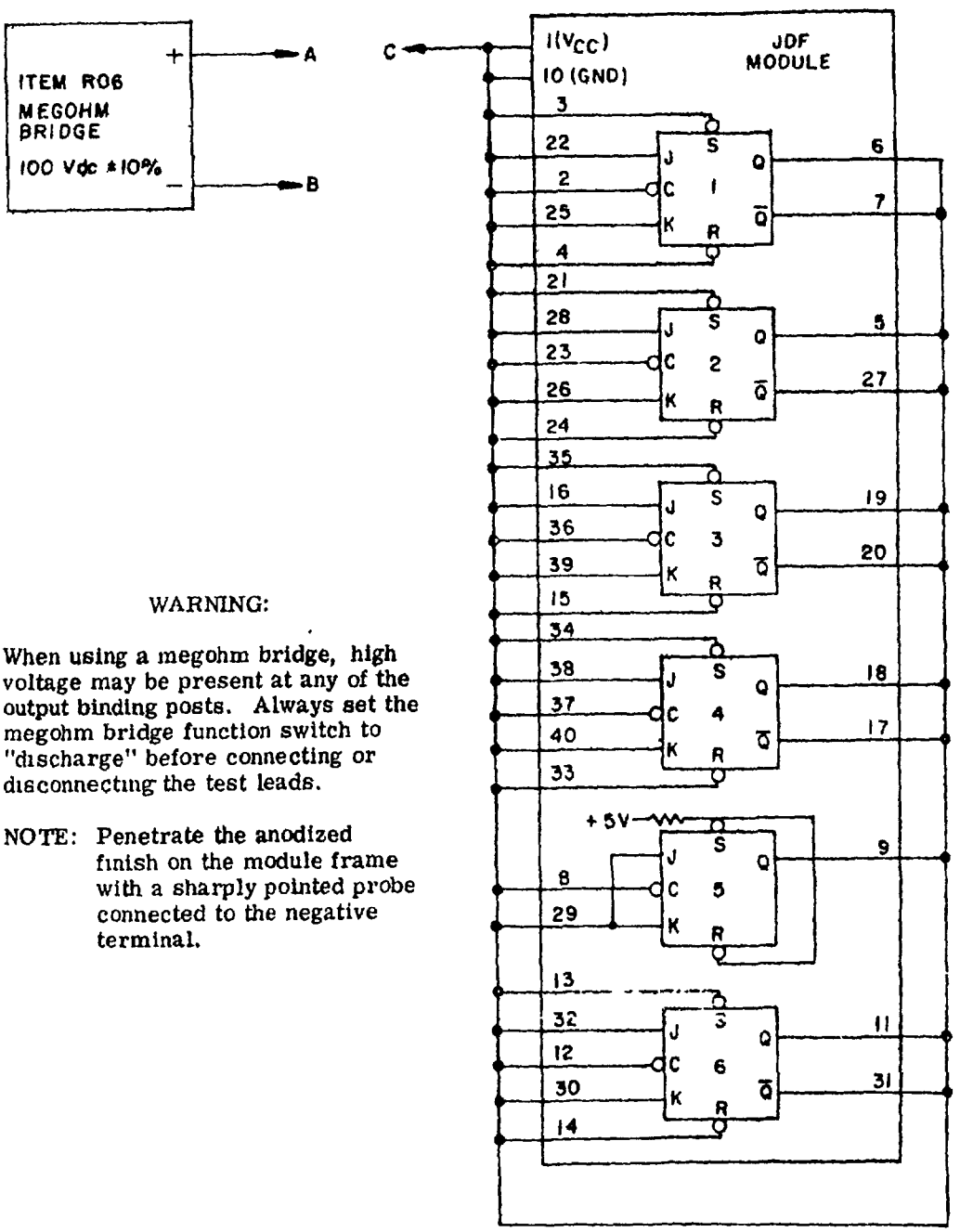
FIGURE 5. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	E to pin	Min	Max		
A	Insulation	C	Frame	10	∞	MΩ	6

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

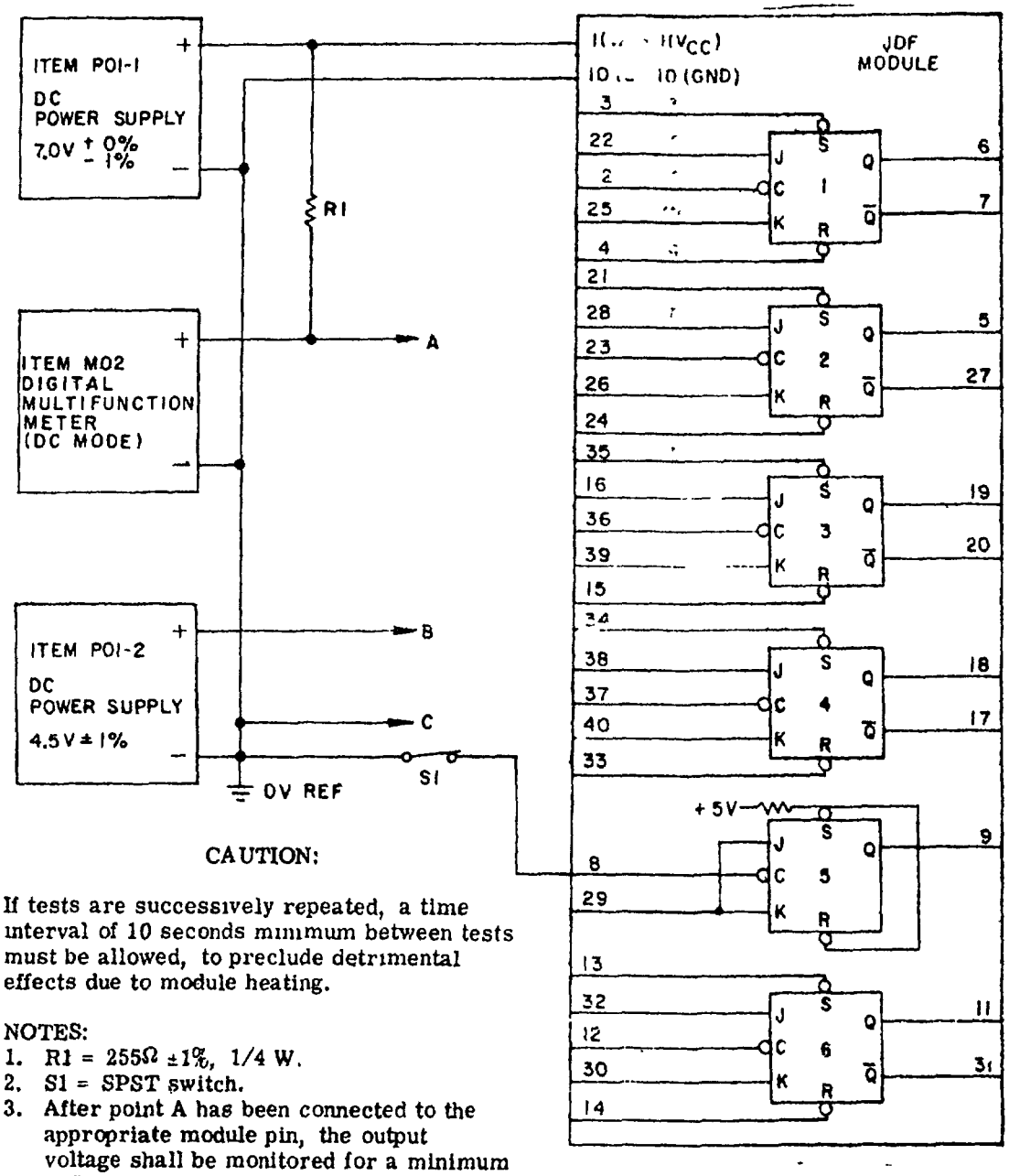
FIGURE 6. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	Q1 V_{OUT} (low loaded)	6	3	4	0.01	0.80	V	7
B2	Q2	5	21	24	↓	↓	↓	↓
B3	Q3	19	35	15				
B4	Q4	18	34	33				
B5	Q5	9	29	Open				
B6	Q6	11	13	14				
B7	$\overline{Q1}$	7	4	3				
B8	$\overline{Q2}$	27	24	21				
B9	$\overline{Q3}$	20	15	35				
B10	$\overline{Q4}$	17	33	34				
B11	$\overline{Q6}$	31	14	13				

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CAUTION:

If tests are successively repeated, a time interval of 10 seconds minimum between tests must be allowed, to preclude detrimental effects due to module heating.

NOTES:
1. R1 = 255Ω ±1%, 1/4 W.
2. S1 = SPST switch.
3. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.
4. For test condition B5 place output Q5 in a low state by opening and closing switch S1 until the low state is obtained

FIGURE 7. V_{OUT} (low-loaded) test circuit for [illegible] table IV.

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TABLE V. Input breakdown current test.

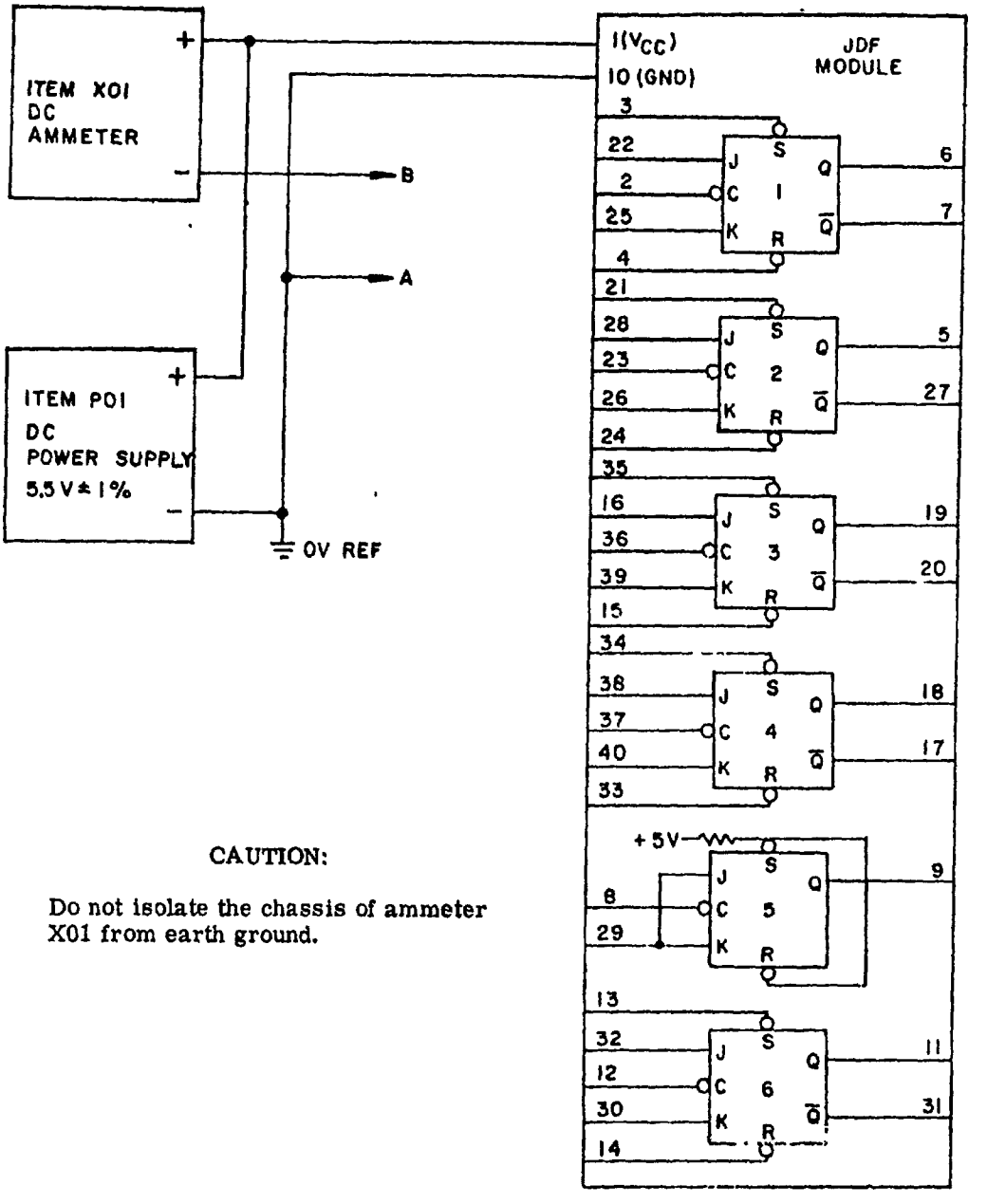
Test condition	Parameter		Connect: A to pin	Connect: B to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
C1	C1	Input breakdown current	3,4,8,12,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	2	0.0	1.0	mA	8
C2	C2		2,3,4,8,12,13,14,15,16,21,22,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	23				
C3	C3		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,37,38,39,40	36				
C4	C4		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,36,38,39,40	37				
C5	C5		2,3,4,12,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	8				
C6	C6		2,3,4,8,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	12				
C7	J1		2,3,4,8,12,13,14,15,16,21,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	22				
C8	J2		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,29,30,32,33,34,35,36,37,38,39,40	28				
C9	J3		2,3,4,8,12,13,14,15,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	16				
C10	J4		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,36,37,39,40	38				
C11	J6		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,33,34,35,36,37,38,39,40	32				
C12	K1		2,3,4,8,12,13,14,15,16,21,22,23,24,26, 28,29,30,32,33,34,35,36,37,38,39,40	25				
C13	K2		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 28,29,30,32,33,34,35,36,37,38,39,40	26				
C14	K3		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,36,37,38,40	39				
C15	K4		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,36,37,38,39	40				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C16	K6	Input breakdown current	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,32,33,34,35,36,37,38,39,40	30	0.0	1.0	mA	8
C17	S1		2,4,8,12,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	3				
C18	S2		2,3,4,8,12,13,14,15,16,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	21				
C19	S3		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,36,37,38,39,40	35				
C20	S4		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,35,36,37,38,39,40	34				
C21	S6		2,3,4,8,12,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	13				
C22	R1		2,3,8,12,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	4				
C23	R2		2,3,4,8,12,13,14,15,16,21,22,23,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	24				
C24	R3		2,3,4,8,12,13,14,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	15				
C25	R4		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,34,35,36,37,38,39,40	33				
C26	R6		2,3,4,8,12,13,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	14				
C27	IK5		2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,30,32,33,34,35,36,37,38,39,40	29		2.0		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

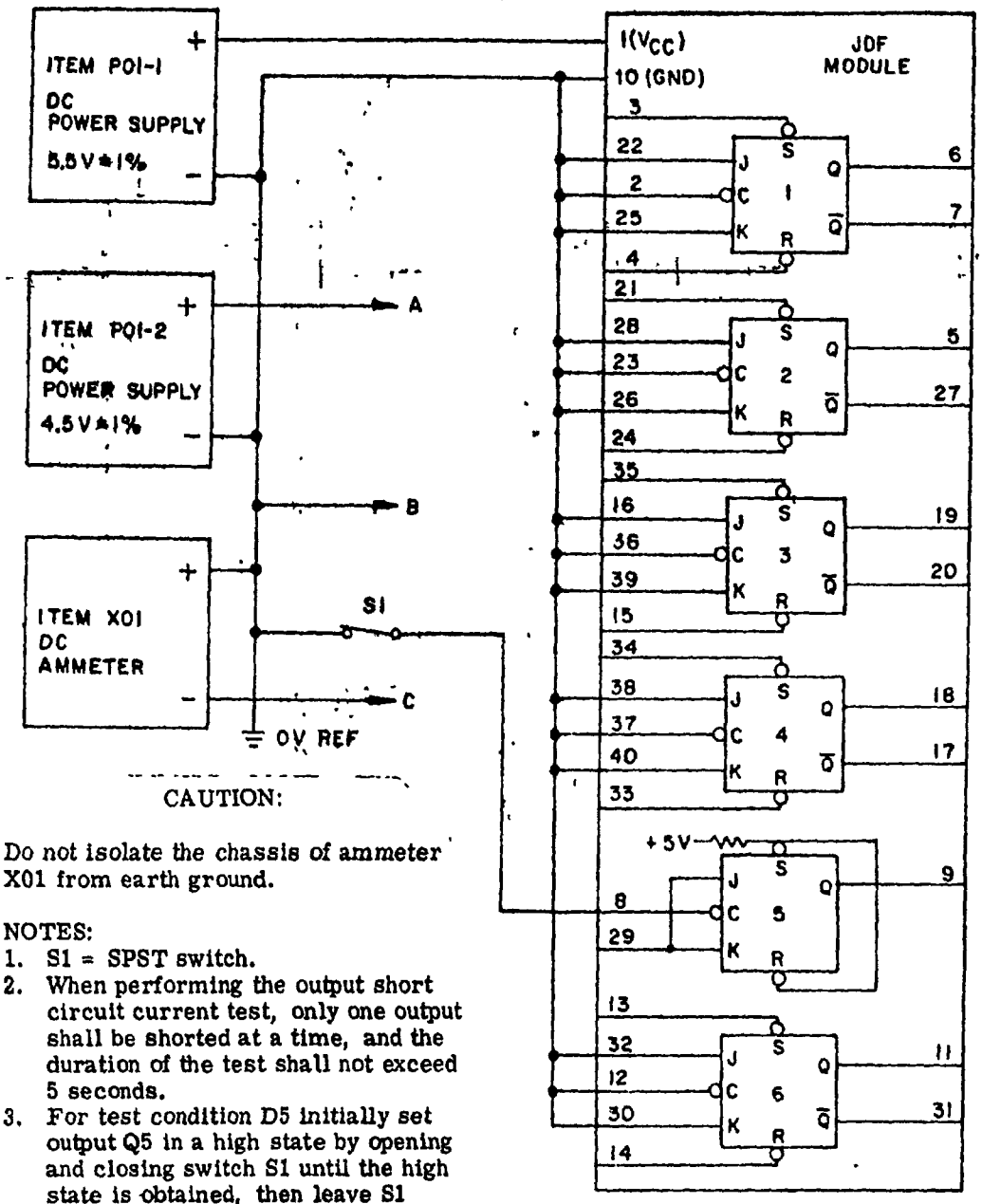
FIGURE 8. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	Q1 Output short circuit current	4	3	6	-40	-100	mA	9
D2	Q2	24	21	5				
D3	Q3	15	35	19				
D4	Q4	33	34	18				
D5	Q5	29	Open	9				
D6	Q6	14	13	11				
D7	$\bar{Q}1$	3	4	7				
D8	$\bar{Q}2$	21	24	27				
D9	$\bar{Q}3$	35	15	20				
D10	$\bar{Q}4$	34	33	17				
D11	Q6	13	14	31				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. S1 = SPST switch.
2. When performing the output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds.
3. For test condition D5 initially set output Q5 in a high state by opening and closing switch S1 until the high state is obtained, then leave S1 closed. Connect point A and make measurement.

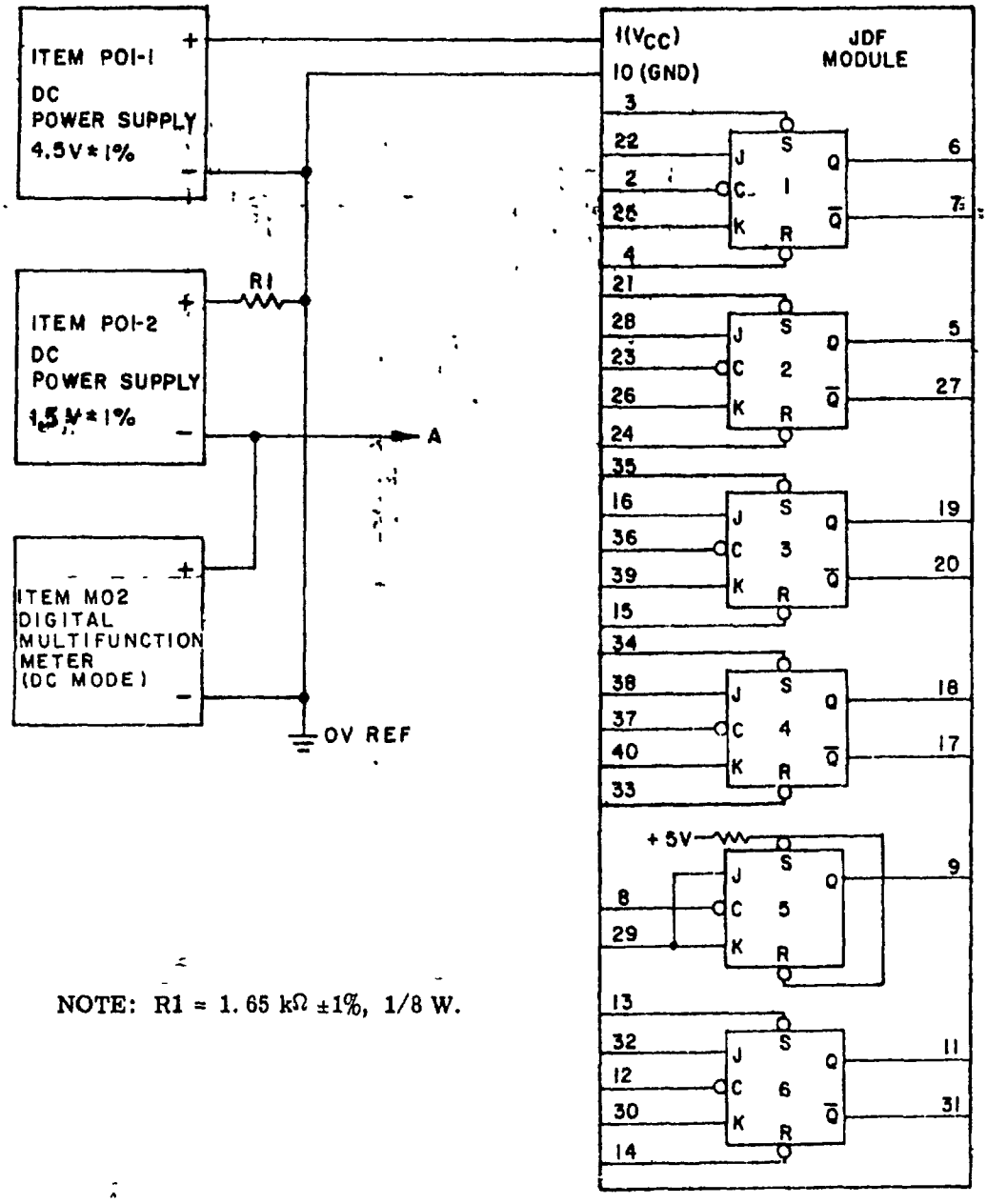
FIGURE 9. Output short circuit current test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	S1 Input clamp diode voltage	3	-0.4	-1.5	V	10
E2	S2	21				
E3	S3	35				
E4	S4	34				
E5	S6	13				
E6	R1	4				
E7	R2	24				
E8	R3	15				
E9	R4	33				
E10	R6	14				
E11	C1	2				
E12	C2	23				
E13	C3	36				
E14	C4	37				
E15	C5	8				
F16	C6	12				
E17	J1	22				
E18	J2	28				
E19	J3	16				
E20	J4	38				
E21	JK5	29				
E22	J6	32				
E23	K1	25				
E24	K2	26				
E25	K3	39				
E26	K4	40				
E27	K6	30				

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NOTE: R1 = 1.65 kΩ ±1%, 1/8 W.

FIGURE 10. Input clamp diode voltage test circuit for table VII.

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TABLE VIII. Functional test.

Test condition 5/	Parameter	Connect			Output logic states											Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Q1	Q2	Q3	Q4	Q5	$\overline{Q6}$	$\overline{Q1}$	$\overline{Q2}$	$\overline{Q3}$	$\overline{Q4}$	$\overline{Q6}$	Min	Max		
F1	Functional (S), J-high, K-low	3,22,21,28,35,16,34,38, 13,32	4,25,24,26,15,39,33,40, 14,30	2/, 3/	H	H	H	H	-	H	L	L	L	L	L	← 1/ →			11
F2	Functional (R), J-high, K-low	4,25,24,26,15,39,33,40, 14,30	3,22,21,28,35,16,34,38, 13,32		L	L	L	L	-	L	H	H	H	H	H				
F3	Functional (J and K), J-high, K-low	25,26,39,40,30	3,4,21,24,35,15,34,33, 13,14,22,28,16,38,32		H	H	H	H	-	H	L	L	L	L	L				
F4	Functional (J and K), J-low, K-high	22,28,16,38,32	3,4,21,24,35,15,34,33, 13,14,25,26,39,40,30		L	L	L	L	-	L	H	H	H	H	H				
F5	Functional (toggle), J-high, K-high	Open	3,4,21,24,35,15,34,33, 13,14,22,25,28,26,16, 39,38,40,32,30		H	H	H	H	-	H	L	L	L	L	L				
F6	Functional (toggle), J-high, K-high	Open	3,4,21,24,35,15,34,33, 13,14,22,25,28,26,16, 39,38,40,32,30		L	L	L	L	-	L	H	H	H	H	H				
F7	Functional (no toggle), J-low, K-low	22,25,28,26,16,39,38, 40,32,30	3,4,21,24,35,15,34,33, 13,14		L	L	L	L	-	L	H	H	H	H	H				
F8	Functional (no toggle), JK-low	29 4/	Open		-	-	-	-	L	-	-	-	-	-	-				
F9	Functional (toggle), JK-high	Open	29		-	-	-	-	H	-	-	-	-	-	-				
F10	Functional (toggle), JK-high	Open	29		-	-	-	-	L	-	-	-	-	-	-				

1/ Output logic states: L = Low-voltage level = 0.0 V to 0.4 V; H = High-voltage level = 2.4 V to 4.5 V.
2/ For test conditions F1-F10 verification of the output logic states in note 1 is obtained by connecting item M0, point C, to each output pin in succession after input conditions have been accomplished.
3/ Q1, Q2, Q3, Q4, Q5, Q6, $\overline{Q1}$, $\overline{Q2}$, $\overline{Q3}$, $\overline{Q4}$, and $\overline{Q6}$ output logic states are measured at output pins 6, 5, 19, 18, 9, 11, 7, 27, 20, 17, and 31 respectively.
4/ Initially connect pin 29 to point B and clock sufficiently to place Q5 in a low state.
5/ Test conditions F1-F10 are to be run in sequence with only the applicable inputs to be changed.

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TABLE IX. Static output voltage (low-level and high-level) test and input threshold check.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	Q1 Static output voltage, low level, and input threshold check	4	3,22,25	6	7	0.0	0.4	V	11
G2	Q2	24	21,28, 26	5	27				
G3	Q3	15	35,16, 39	19	20				
G4	Q4	33	34,38, 40	18	17				
G5 1/	Q5	Open	29	9	Open				
G6	Q6	14	13,32, 30	11	31				
G7	$\overline{Q1}$	3	4,22,25	7	6				
G8	$\overline{Q2}$	21	24,28, 26	27	5				
G9	$\overline{Q3}$	35	15,16, 39	20	19				
G10	$\overline{Q4}$	34	33,38, 40	17	18				
G11	$\overline{Q6}$	13	14,32, 30	31	11				
H1	Q1 Static output voltage, high level, and input threshold check	3,22,25	4	6	7	2.4	4.5		
H2	Q2	21,28, 26	24	5	27				
H3	Q3	35,16, 39	15	19	20				
H4	Q4	34,38, 40	33	18	17				

See footnotes at end of table.

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TABLE IX. Static output voltage (low-level and high-level) test and input threshold check - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H5 2/	Q5 Static output voltage, high level, and input threshold check	29	Open	9	Open	2.4	4.5	V	11
H6	Q6	13,32, 30	14	11	31				
H7	$\bar{Q}1$	4,22,25	3	7	6				
H8	$\bar{Q}2$	24,28, 26	21	27	5				
H9	$\bar{Q}3$	15,16, 39	35	20	19				
H10	$\bar{Q}4$	33,38, 40	34	17	18				
H11	$\bar{Q}6$	14,32, 30	13	31	11				

1/ For test condition G5, place Q5 in a low state by providing sufficient clock pulses to place the output in a low state.

2/ For test condition H5, place Q5 in a high state by providing sufficient clock pulses to place the output in a high state.

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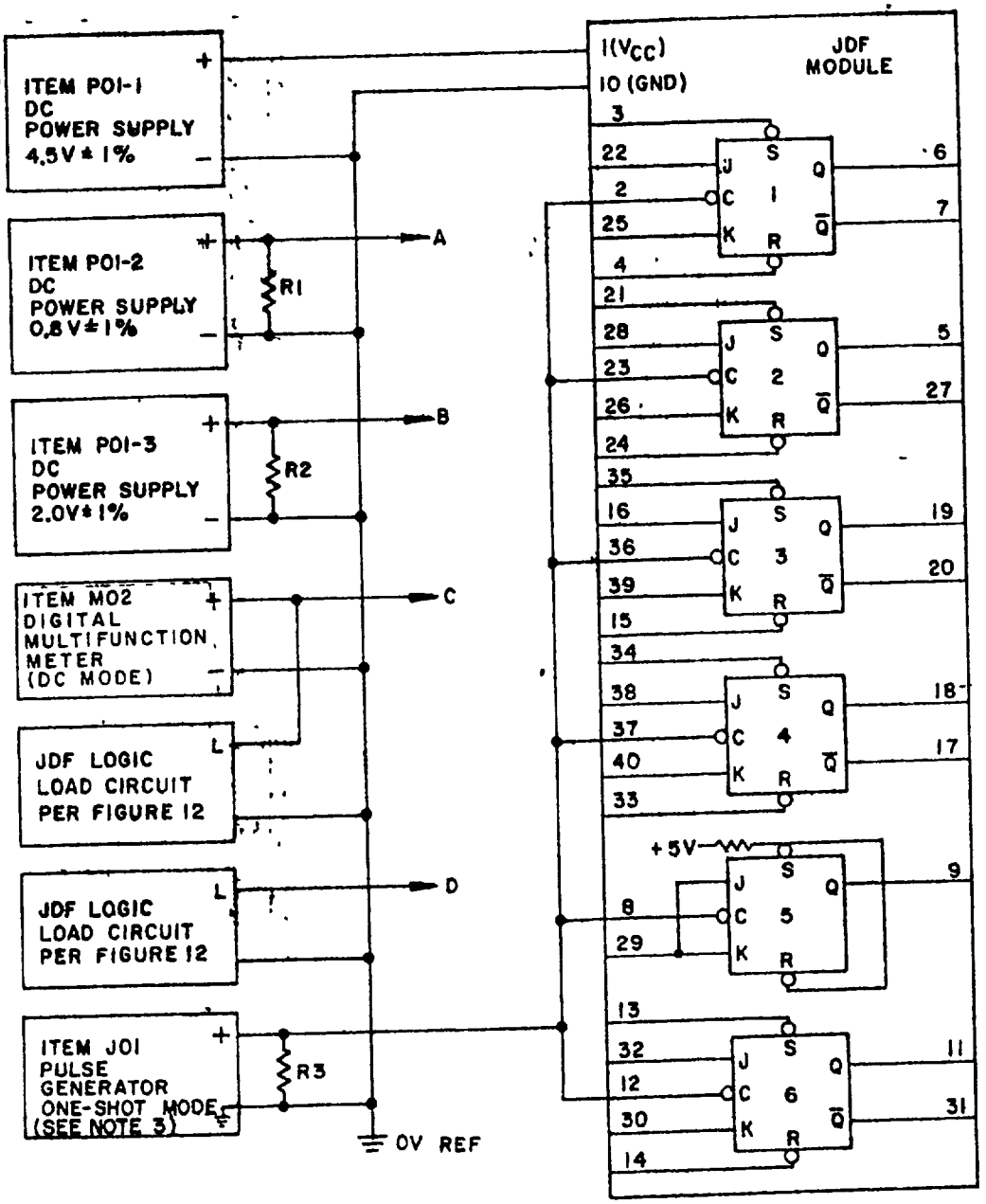
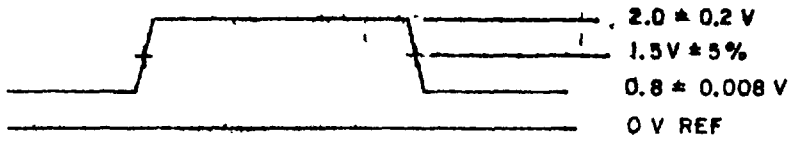


FIGURE 11. Functional, static output voltage (low-level and high-level) and input threshold check test circuit for tables VIII and IX.

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Check the source to verify that this is the current version before use.

NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load circuits are not required for functional test conditions F1-F10.
3. Pulse generator setup:
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PRR = 1.0 MHz ±10 Hz.

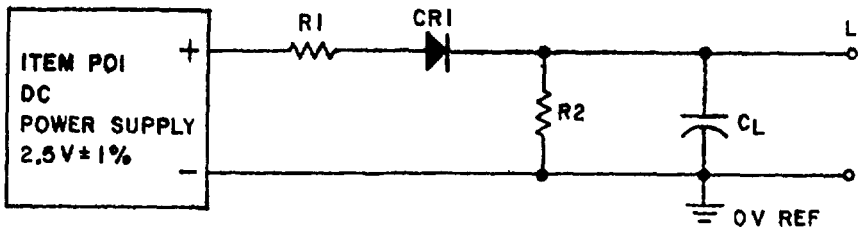


The PRR is used for setup procedures only. When the pulse has been setup, the pulse generator should be placed in the one-shot mode.

4. After the input conditions are setup per each test condition, apply one pulse using the J01 pulse generator, then make the required measurement.

FIGURE 11. Functional, static output voltage (low-level and high-level) and input threshold check test circuit for tables VIII and IX - Continued.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 16.0 mA ±5% with point L held at 0.4 V.
3. R2 = 4.75 kΩ ±1%, 1/8 W.
4. For test conditions M1-M11, N1-N11, P1-P11, Q1-Q11, R1-R10, S1-S10, T1-T10, U1-U11, V1-V6, W1-W6, X1-X10, and Y1-Y10, C_L = 50 ±5 pF (including probe and parasitic capacitance).
5. For test conditions M12-M22, N12-N22, P12-P22, Q12-Q22, R11-R20, S11-S20, and T11-T20, C_L = 150 ±5 pF (including probe and parasitic capacitance).
6. Capacitor C_L is not required for dc electrical tests G and H.
7. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

FIGURE 12. JDF logic load circuit.

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TABLE X. Input current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	J1	Input current, high level	22	2,3,4,8,12,13,14,15,16,21,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	0	50	µA	13
J2	J2		28	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,29,30,32,33,34,35,36,37,38,39,40				
J3	J3		16	2,3,4,8,12,13,14,15,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40				
J4	J4		38	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,36,37,39,40				
J5	J6		32	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,33,34,35,36,37,38,39,40				
J6	[illegible]		25	2,3,4,8,12,13,14,15,16,21,22,23,24,26, 28,29,30,32,33,34,35,36,37,38,39,40				
J7	K2		26	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 28,29,30,32,33,34,35,36,37,38,39,40				
J8	K3		39	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,36,37,38,40				
J9	K4		40	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,36,37,38,39				
J10	[illegible]		30	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,32,33,34,35,36,37,38,39,40				
J11	S1		3	2,4,8,12,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40		100		
J12	S2		21	2,3,4,8,12,13,14,15,16,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40				
J13	S3		35	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,36,37,38,39,40				
J14	S4		34	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,35,36,37,38,39,40				
J15	S6		13	2,3,4,8,12,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40				

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J16	R1	Input current, high level	4	2,3,8,12,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	0	100	μA	13
J17	R2		24	2,3,4,8,12,13,14,15,16,21,22,23,25,26, 28,29,30,32,33,34,35,36,37,38,39,40				
J18	R3		15	2,3,4,8,12,13,14,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40				
J19	R4		33	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,34,35,36,37,38,39,40				
J20	R6		14	2,3,4,8,12,13,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40				
J21	JK5		29	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,30,32,33,34,35,36,37,38,39,40				
J22	C1		2	3,4,8,12,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40	-1.0	0.2	mA	
J23	C2		23	2,3,4,8,12,13,14,15,16,21,22,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40				
J24	C3		36	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,37,38,39,40				
J25	C4		37	2,3,4,8,12,13,14,15,16,21,22,23,24,25, 26,28,29,30,32,33,34,35,36,38,39,40				
J26	C5		8	2,3,4,12,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40				
J27	C6		12	2,3,4,8,13,14,15,16,21,22,23,24,25,26, 28,29,30,32,33,34,35,36,37,38,39,40				

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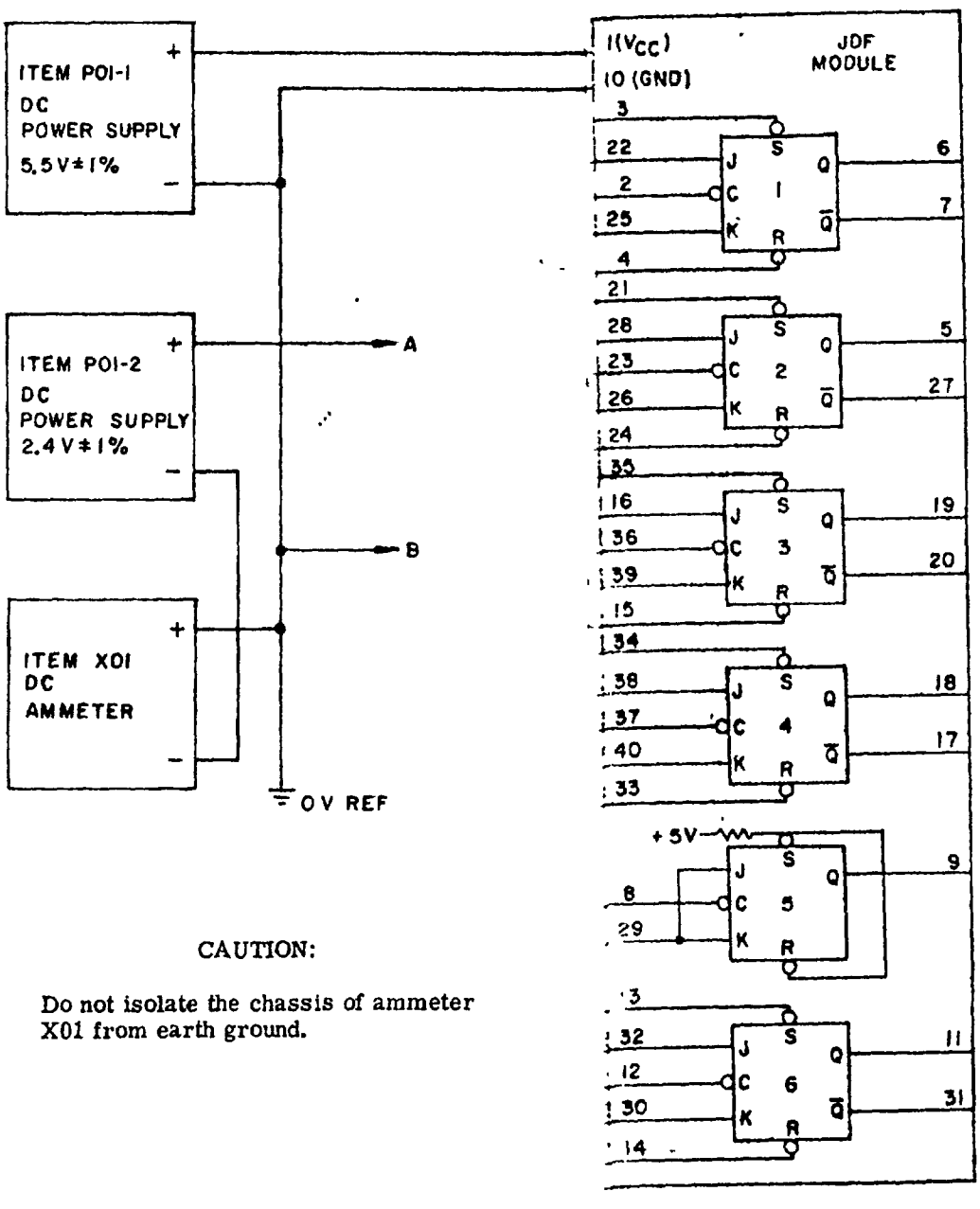


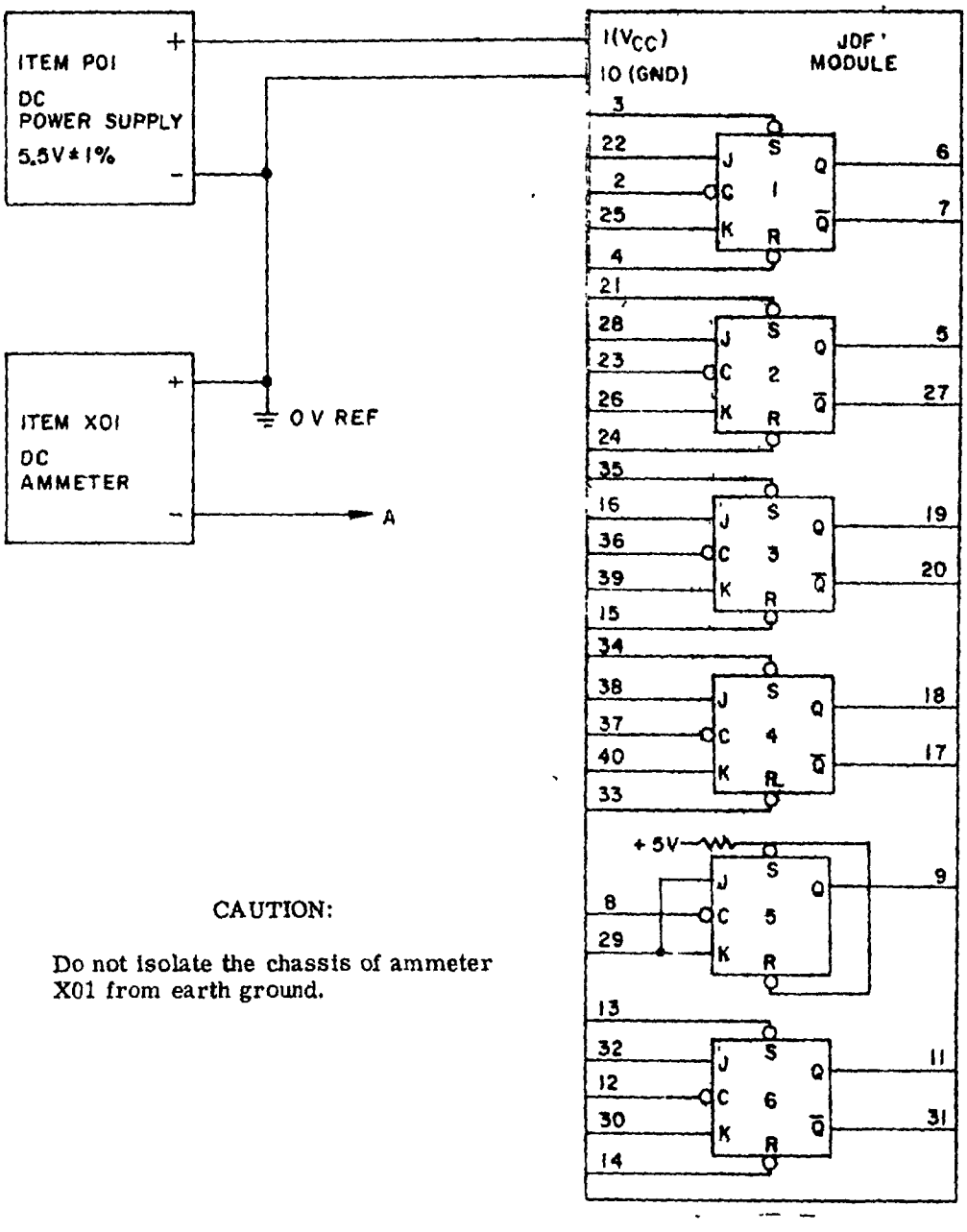
FIGURE 13. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	J1 Input current, low level	22	-0.1	-2.2	mA	14
K2	J2	28				
K3	J3	16				
K4	J4	38				
K5	J6	32				
K6	K1	25				
K7	K2	26				
K8	K3	39				
K9	K4	40				
K10	K6	30				
K11	S1	3	-0.2	-4.4		
K12	S2	21				
K13	S3	35				
K14	S4	34				
K15	S6	13				
K16	R1	4				
K17	R2	24				
K18	R3	15				
K19	R4	33				
K20	R6	14				
K21	JK5	29				
K22	C1	2	-0.3	-6.9		
K23	C2	23				
K24	C3	36				
K25	C4	37				
K26	C5	8				
K27	C6	12				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 14. Input current (low-level) test circuit for table XI.

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TABLE XII. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current	30	228	mA	15

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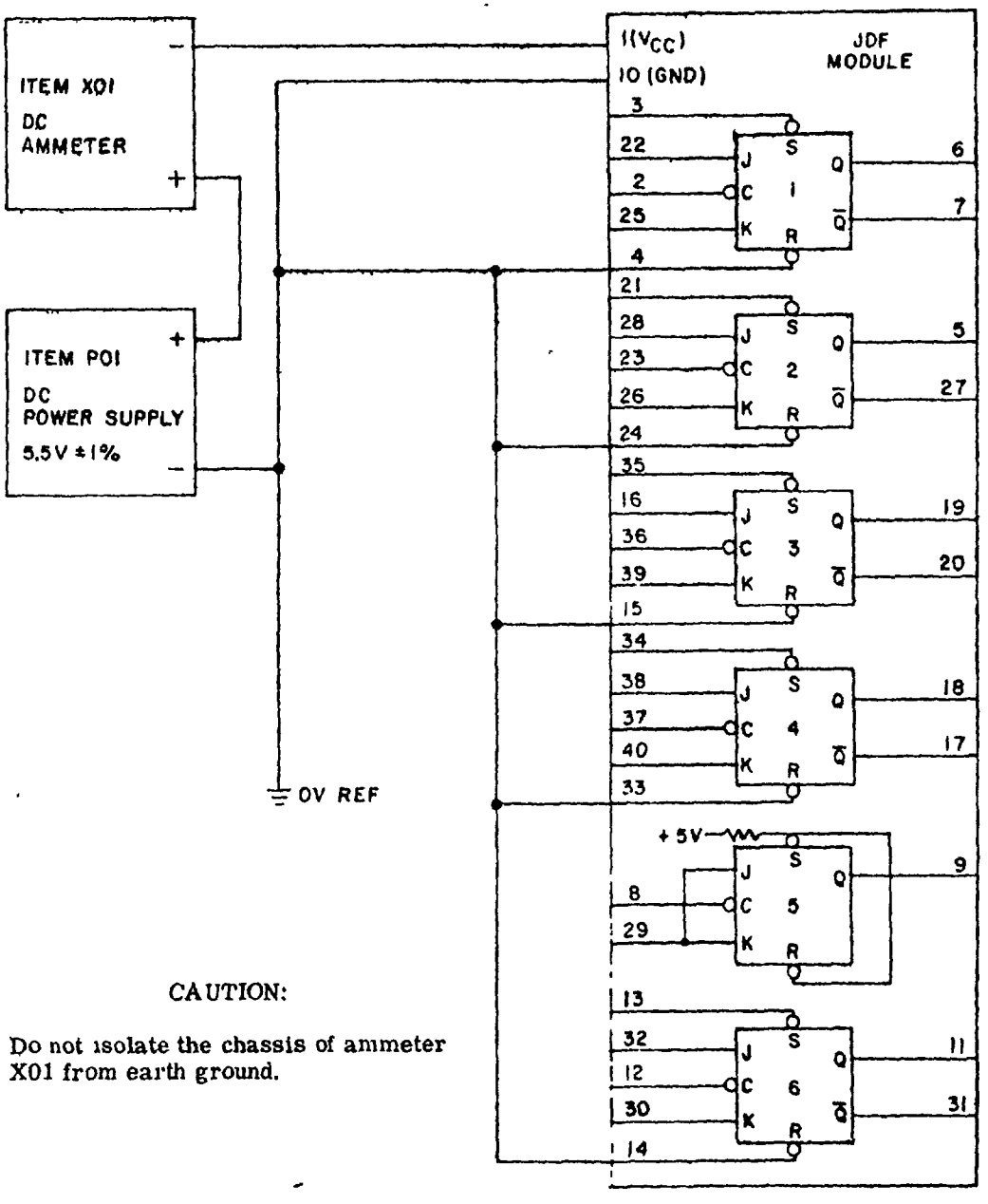


FIGURE 15. Power supply current test circuit for table XII.

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TABLE XIII. Transition time and propagation delay time (clock) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	Q1 Transition time (t_{TLH}), C_L = 50 pF	3,22,25,4	2	6	7	3	20	ns	16, 17
M2	Q2	21,28,26,24	23	5	27				
M3	Q3	35,16,39,15	36	19	20				
M4	Q4	34,38,40,33	37	18	17				
M5	Q5	29	8	9	Open				
M6	Q6	13,32,30,14	12	11	31				
M7	$\overline{Q1}$	3,22,25,4	2	7	6				
M8	$\overline{Q2}$	21,28,26,24	23	27	5				
M9	$\overline{Q3}$	35,16,39,15	36	20	19				
M10	$\overline{Q4}$	34,38,40,33	37	17	18				
M11	$\overline{Q6}$	13,32,30,14	12	31	11				
M12	Q1 Transition time (t_{TLH}), C_L = 150 pF	3,22,25,4	2	6	7	5	30		
M13	Q2	21,28,26,24	23	5	27				
M14	Q3	35,16,39,15	36	19	20				
M15	Q4	34,38,40,33	37	18	17				
M16	Q5	29	8	9	Open				
M17	Q6	13,32,30,14	12	11	31				
M18	$\overline{Q1}$	3,22,25,4	2	7	6				
M19	$\overline{Q2}$	21,28,26,24	23	27	5				
M20	$\overline{Q3}$	35,16,39,15	36	20	19				
M21	$\overline{Q4}$	34,38,40,33	37	17	18				
M22	$\overline{Q6}$	13,32,30,14	12	31	11				

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TABLE XIII. Transition time and propagation delay time (clock) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Q1 Transition time (t_{THL}), C_L = 50 pF	3,22,25,4	2	6	7	3	20	ns	16, 17
N2	Q2	21,28,26,24	23	5	27				
N3	Q3	35,16,39,15	36	19	20				
N4	Q4	34,38,40,33	37	18	17				
N5	Q5	29	8	9	Open				
N6	Q6	13,32,30,14	12	11	31				
N7	$\overline{Q1}$	3,22,25,4	2	7	6				
N8	$\overline{Q2}$	21,28,26,24	23	27	5				
N9	$\overline{Q3}$	35,16,39,15	36	20	19				
N10	$\overline{Q4}$	34,38,40,33	37	17	18				
N11	$\overline{Q6}$	13,32,30,14	12	31	11				
N12	Q1 Transition time (t_{THL}), C_L = 150 pF	3,22,25,4	2	6	7	5	25		
N13	Q2	21,28,26,24	23	5	27				
N14	Q3	35,16,39,15	36	19	20				
N15	Q4	34,38,40,33	37	18	17				
N16	Q5	29	8	9	Open				
N17	Q6	13,32,30,14	12	11	31				
N18	$\overline{Q1}$	3,22,25,4	2	7	6				
N19	$\overline{Q2}$	21,28,26,24	23	27	5				
N20	$\overline{Q3}$	35,16,39,15	36	20	19				
N21	$\overline{Q4}$	34,38,40,33	37	17	18				
N22	$\overline{Q6}$	13,32,30,14	12	31	11				

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TABLE XIII. Transition time and propagation delay time (clock) test - Continued.

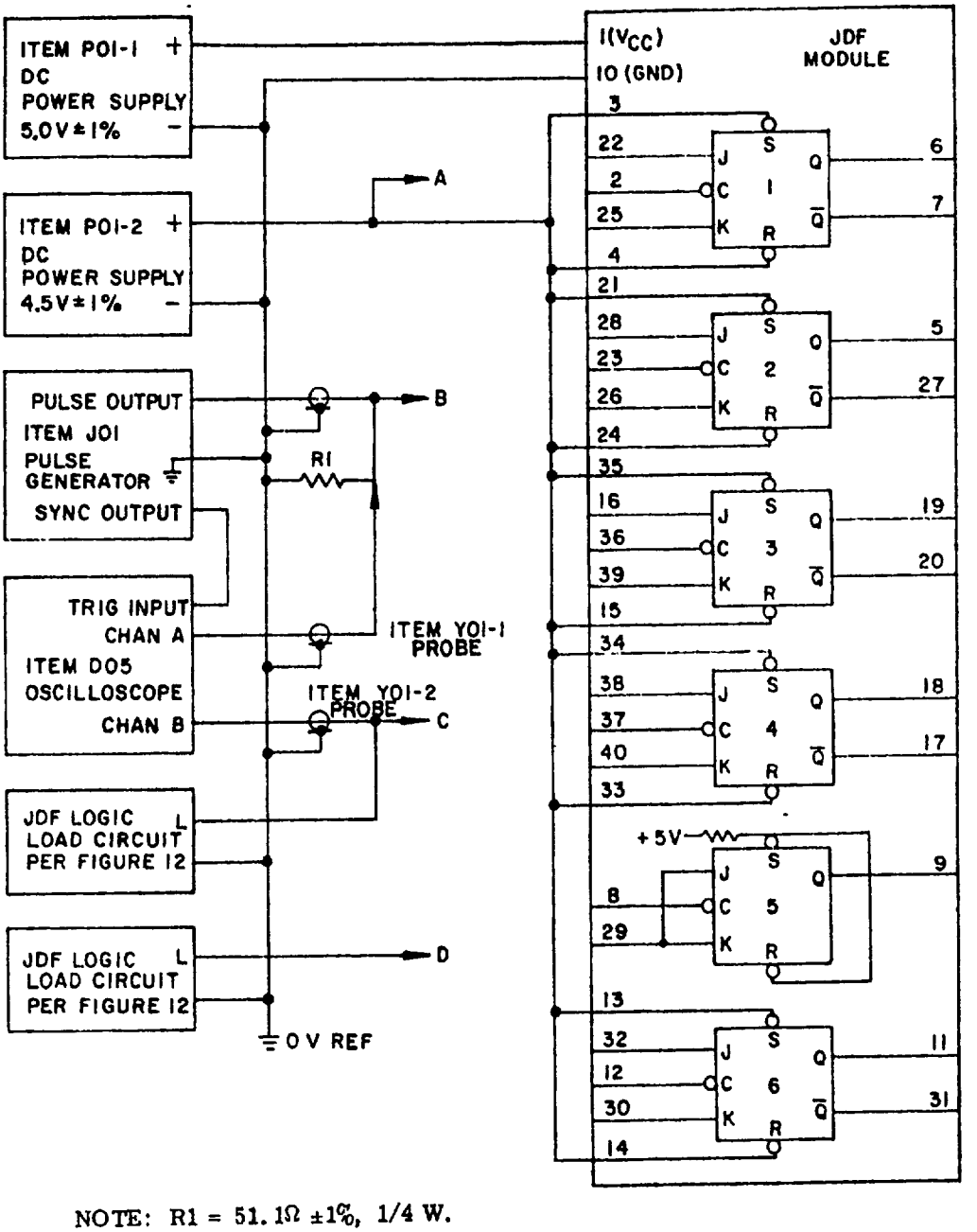
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	Q1 Propagation delay time (t_{PLH}), clock, C_L = 50 pF	3,22,25,4	2	6	7	5	18	ns	16, 17
P2	Q2	21,28,26,24	23	5	27				
P3	Q3	35,16,39,15	36	19	20				
P4	Q4	34,38,40,33	37	18	17				
P5	Q5	29	8	9	Open				
P6	Q6	13,32,30,14	12	11	31				
P7	$\overline{Q1}$	3,22,25,4	2	7	6				
P8	$\overline{Q2}$	21,28,26,24	23	27	5				
P9	$\overline{Q3}$	35,16,39,15	36	20	19				
P10	$\overline{Q4}$	34,38,40,33	37	17	18				
P11	$\overline{Q6}$	13,32,30,14	12	31	11				
P12	Q1 Propagation delay time (t_{PLH}), clock, C_L = 150 pF	3,22,25,4	2	6	7		28		
P13	Q2	21,28,26,24	23	5	27				
P14	Q3	35,16,39,15	36	19	20				
P15	Q4	34,38,40,33	37	18	17				
P16	Q5	29	8	9	Open				
P17	Q6	13,32,30,14	12	11	31				
P18	$\overline{Q1}$	3,22,25,4	2	7	6				
P19	$\overline{Q2}$	21,26,26,24	23	27	5				
P20	$\overline{Q3}$	35,16,39,15	36	20	19				
P21	$\overline{Q4}$	34,38,40,33	37	17	18				
P22	$\overline{Q6}$	13,32,30,14	12	31	11				

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TABLE XIII. Transition time and propagation delay time (clock) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	Q1 Propagation delay time (t_{PHL}), clock, C_L = 50 pF	3,22,25,4	2	6	7	5	27	ns	16, 17
Q2	Q2	21,28,26,24	23	5	27				
Q3	Q3	35,16,39,15	36	19	20				
Q4	Q4	34,38,40,33	37	18	17				
Q5	Q5	29	8	9	Open				
Q6	Q6	13,32,30,14	12	11	31				
Q7	$\overline{Q1}$	3,22,25,4	2	7	6				
Q8	$\overline{Q2}$	21,28,26,24	23	27	5				
Q9	$\overline{Q3}$	35,16,39,15	36	20	19				
Q10	$\overline{Q4}$	34,38,40,33	37	17	18				
Q11	$\overline{Q6}$	13,32,30,14	12	31	11				
Q12	Q1 Propagation delay time (t_{PHL}), clock, C_L = 150 pF	3,22,25,4	2	6	7		37		
Q13	Q2	21,28,26,24	23	5	27				
Q14	Q3	35,16,39,15	36	19	20				
Q15	Q4	34,38,40,33	37	18	17				
Q16	Q5	29	8	9	Open				
Q17	Q6	13,32,30,14	12	11	31				
Q18	$\overline{Q1}$	3,22,25,4	2	7	6				
Q19	$\overline{Q2}$	21,28,26,24	23	27	5				
Q20	$\overline{Q3}$	35,16,39,15	36	20	19				
Q21	$\overline{Q4}$	34,38,40,33	37	17	18				
Q22	$\overline{Q6}$	13,32,30,14	12	31	11				

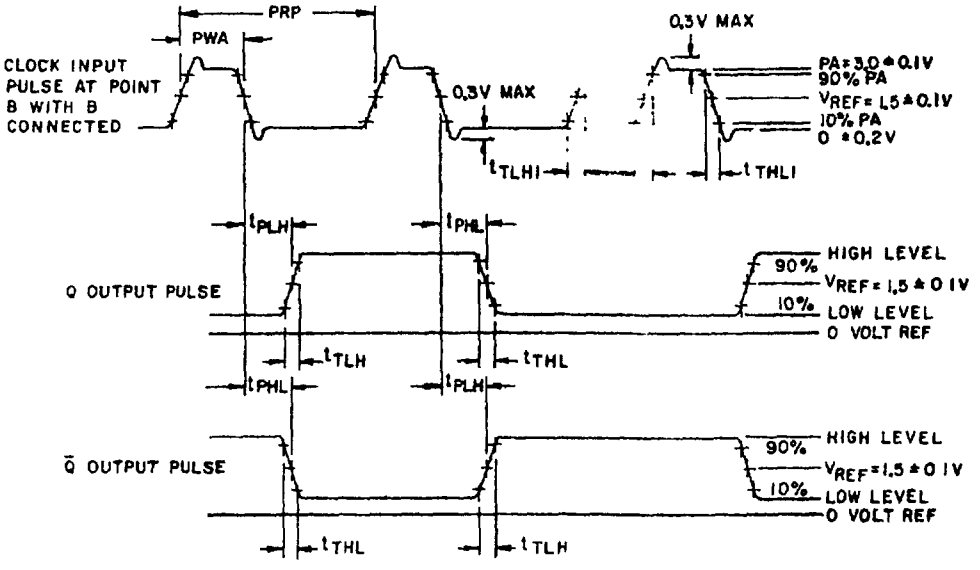
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 16. Transition time and propagation delay time (clock) test circuit for table XIII.

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NOTES:

1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 250 ±50 ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns.
2. Verify that all outputs have a high level $\geq$ 2.4 V v and a low level $\leq$ 0.4 V before making measurements per table XIII.
3. All voltages are referenced to ground.
4. Automatic tester setup procedure:
   (a) Circuits 1, 2, 3, 4, and 6 can be initially setup by grounding the set or reset input pins before connecting point B
   (b) Circuit 5 can be initially setup by pulsing the c ne clock input to obtain the desired output state before connecting point B.

FIGURE 17. Transition time and propagation delay time (clock) test circuit waveforms for table XIII.

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TABLE XIV. Propagation delay time (set and reset) test.

Test condition	Parameter	Connect						Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	G to pin	Min	Max		
R1	Q1 Propagation delay time (t_{PLH}), set, C = low, C_L = 50 pF	3,F	4	6	7	22,2,25	Open	5	15	ns	18, 19
R2	Q2	21,F	24	5	27	28,23,26					
R3	Q3	35,F	15	19	20	16,36,39					
R4	Q4	34,F	33	18	17	38,37,40					
R5	Q6	13,F	14	11	31	32,12,30					
R6	$\overline{Q1}$ Propagation delay time (t_{PLH}), reset, C = low, C_L = 50 pF	3	4,F	7	6	22,2,25					
R7	$\overline{Q2}$	21	24,F	27	5	28,23,26					
R8	$\overline{Q3}$	35	15,F	20	19	16,36,39					
R9	$\overline{Q4}$	34	33,F	17	18	38,37,40					
R10	$\overline{Q6}$	13	14,F	31	11	32,12,30					
R11	Q1 Propagation delay time (t_{PLH}), set, C = low, C_L = 150 pF	3,F	4	6	7	22,2,25			25		
R12	Q2	21,F	24	5	27	28,23,26					
R13	Q3	35,F	15	19	20	16,36,39					
R14	Q4	34,F	33	18	17	38,37,40					
R15	Q6	13,F	14	11	31	32,12,30					
R16	$\overline{Q1}$ Propagation delay time (t_{PLH}), reset, C = low, C_L = 150 pF	3	4,F	7	6	22,2,25					
R17	$\overline{Q2}$	21	24,F	27	5	28,23,26					
R18	$\overline{Q3}$	35	15,F	20	19	16,36,39					
R19	$\overline{Q4}$	34	33,F	17	18	38,37,40					
R20	$\overline{Q6}$	13	14,F	31	11	32,12,30					

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TABLE XIV. Propagation delay time (set and reset) test - Continued.

Test condition	Parameter	Connect						Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	G to pin	Min	Max		
S1	$\overline{Q}1$ Propagation delay time (t_{PHL}), set, C = low, C_L = 50 pF	3,F	4	7	6	22,2,25	Open	5	38	ns	18, 19
S2	$\overline{Q}2$	21,F	24	27	5	28,23,26					
S3	$\overline{Q}3$	35,F	15	20	19	16,36,39					
S4	$\overline{Q}4$	34,F	33	17	18	38,37,40					
S5	$\overline{Q}6$	13,F	14	31	11	32,12,30					
S6	Q1 Propagation delay time (t_{PHL}), reset, C = low, C_L = 50 pF	3	4,F	6	7	22,2,25					
S7	Q2	21	24,F	5	27	28,23,26					
S8	Q3	35	15,F	19	20	16,36,39					
S9	Q4	34	33,F	18	17	38,37,40					
S10	Q6	13	14,F	11	31	32,12,30					
S11	$\overline{Q}1$ Propagation delay time (t_{PHL}), set, C = low, C_L = 150 pF	3,F	4	7	6	22,2,25			48		
S12	$\overline{Q}2$	21,F	24	27	5	28,23,26					
S13	$\overline{Q}3$	35,F	15	20	19	16,36,39					
S14	$\overline{Q}4$	34,F	33	17	18	38,37,40					
S15	$\overline{Q}6$	13,F	14	31	11	32,12,30					
S16	Q1 Propagation delay time (t_{PHL}), reset, C = low, C_L = 150 pF	3	4,F	6	7	22,2,25					
S17	Q2	21	24,F	5	27	28,23,26					
S18	Q3	35	15,F	19	20	16,36,39					
S19	Q4	34	33,F	18	17	38,37,40					
S20	Q6	13	14,F	11	31	32,12,30					

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TABLE XIV. Propagation delay time (set and reset) test - Continued.

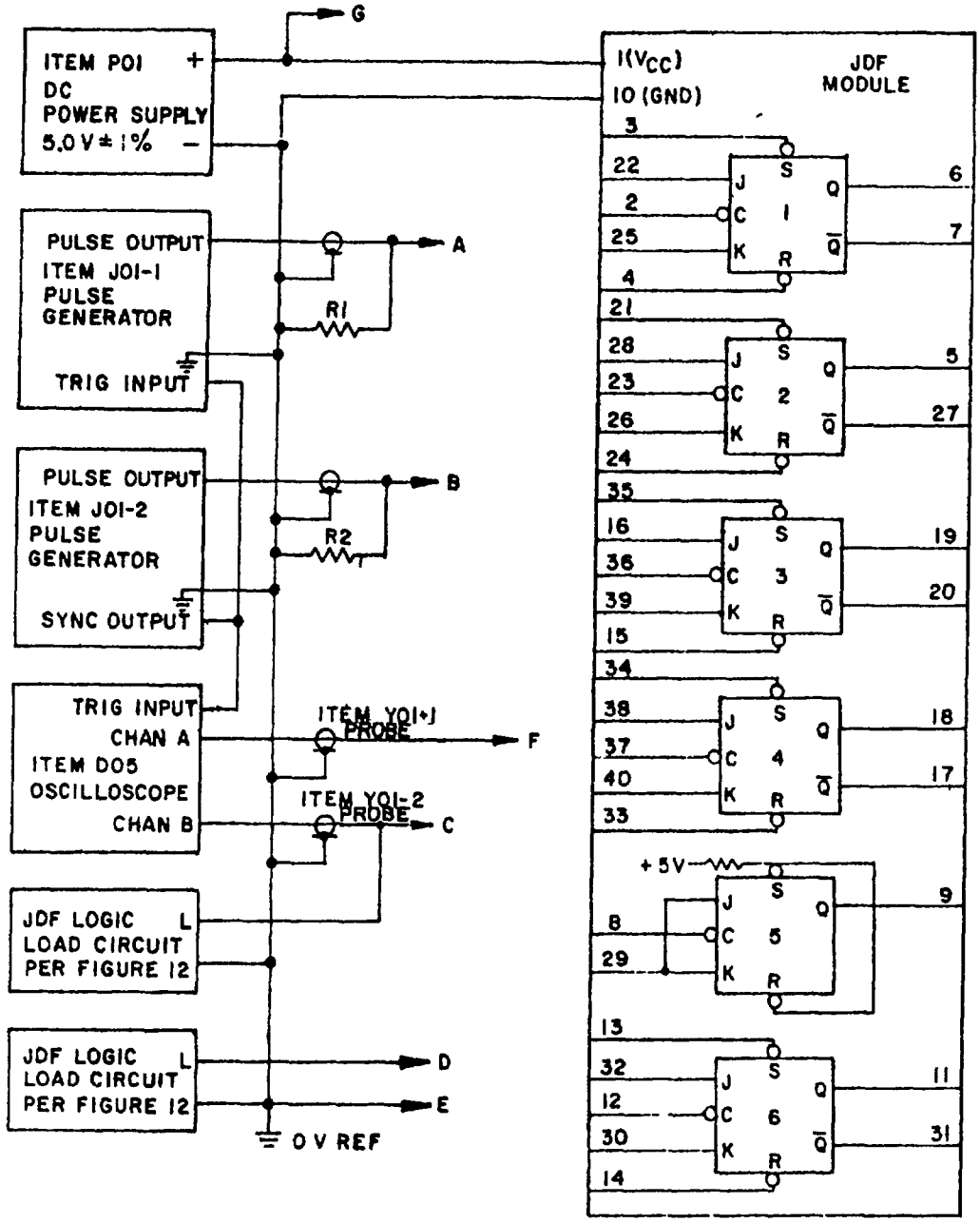
Test condition	Parameter	Connect						Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	G to pin	Min	Max		
T1	$\overline{Q}1$ Propagation delay time (t_{PHL}), set, C = high, C_L = 50 pF	3,F	4	7	6	Open	22,2, 25	5	28	ns	18, 19
T2	Q2	21,F	24	27	5		28, 23,26				
T3	$\overline{Q}3$	35,F	15	20	19		16, 36,39				
T4	$\overline{Q}4$	34,F	33	17	18		38, 37,40				
T5	$\overline{Q}6$	13,F	14	31	11		32, 12,30				
T6	Q1 Propagation delay time (t_{PHL}), reset, C = high, C_L = 50 pF	3	4,F	6	7		22,2, 25				
T7	Q2	21	24,F	5	27		28, 23,26				
T8	Q3	35	15,F	19	20		16, 36,39				
T9	Q4	34	33,F	18	17		38, 37,40				
T10	Q6	13	14,F	11	31		32, 12,30				
T11	$\overline{Q}1$ Propagation delay time (t_{PHL}), set, C = high, C_L = 150 pF	3,F	4	7	6		22,2, 25		38		
T12	$\overline{Q}2$	21,F	24	27	5		28, 23,26				
T13	Q3	35,F	15	20	19		16, 36,39				
T14	$\overline{Q}4$	34,F	33	17	18		38, 37,40				
T15	$\overline{Q}6$	13,F	14	31	11		32, 12,30				

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TABLE XIV. Propagation delay time (set and reset) test - Continued.

Test condition	Parameter	Connect						Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	G to pin	Min	Max		
T16	Q1 Propagation delay time (t_{PHL}), reset, C = high, C_L = 150 pF	3	4,F	6	7	Open	22,2, 25	5	38	ns	18, 19
T17	Q2	21	24,F	5	27		28, 23,26				
T18	Q3	35	15,F	19	20		16, 36,39				
T19	Q4	34	33,F	18	17		38, 37,40				
T20	Q6	13	14,F	11	31		32, 12,30				

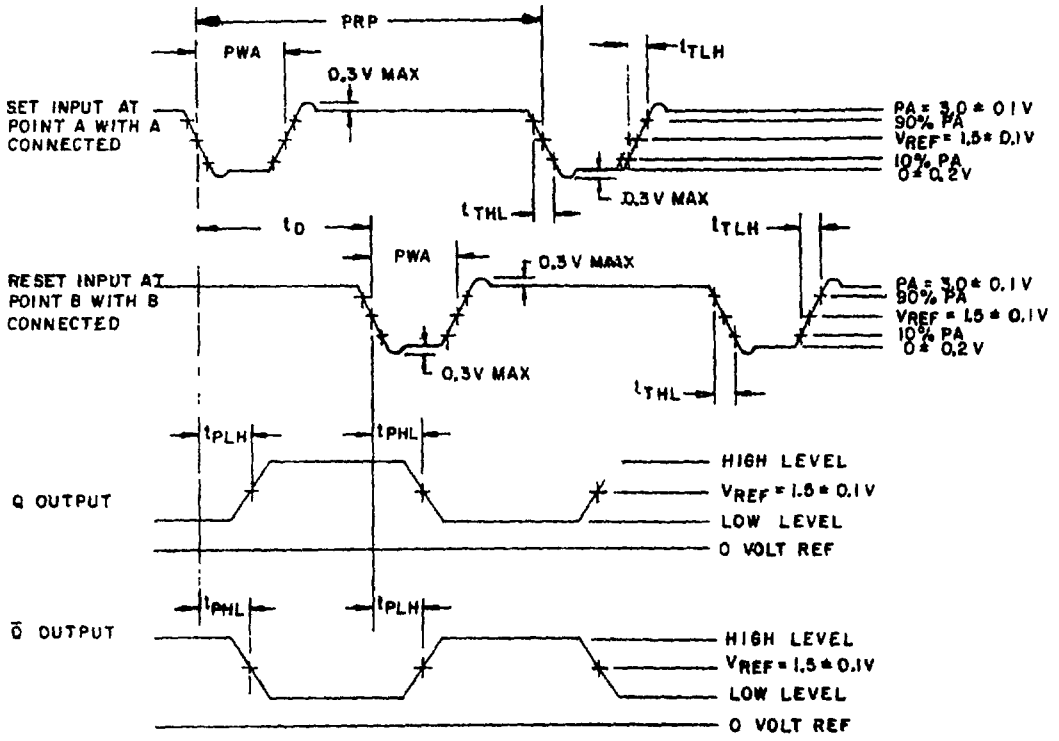
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NOTE: R1 = R2 = 51.1Ω ±1%, 1/4 W.

FIGURE 18. Propagation delay time (set and reset) test circuit for table XIV.

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NOTES:
1. Input pulse parameters:
   (a) PWA = 21 ±1 ns (minimum set and reset pulse width).
   (b) PRP = 500 ±50 ns.
   (c) t_D = 250 ±25 ns.
   (d) $t_{TLH} = t_{THL}$ = 6 ±2 ns.
2. Pulse width (PWA) requirement shall be verified before making measurements.
3. Verify that all outputs have a high $\geq$ 2.4 V and a low $\leq$ 0.4 V before making measurements.
4. All voltages are referenced to ground.

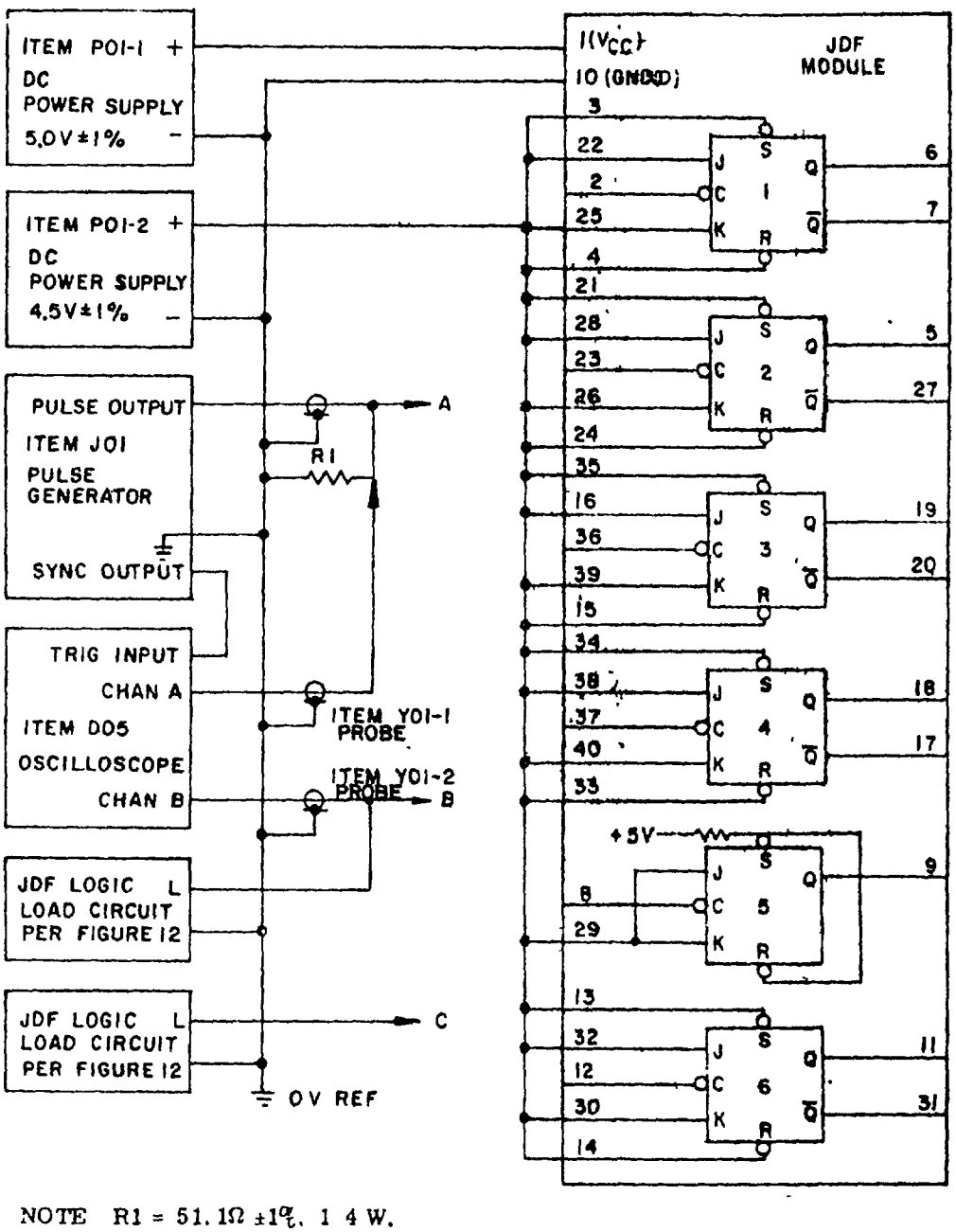
FIGURE 19. Propagation delay time (set and reset) test circuit waveforms for table XIV.

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TABLE XV. Maximum operating frequency and minimum clock pulse width check.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U1	Q1 Maximum operating frequency and minimum clock pulse width (PW measurement)	2	6	7	27	50	ns	20, 21
U2	Q2	23	5	27				
U3	Q3	36	19	20				
U4	Q4	37	18	17				
U5	Q5	8	9	Open				
U6	Q6	12	11	31				
U7	$\overline{Q1}$ Maximum operating frequency and minimum clock pulse width (PRP measurement)	2	7	6	54	66		
U8	$\overline{Q2}$	23	27	5				
U9	$\overline{Q3}$	36	20	19				
U10	$\overline{Q4}$	37	17	18				
U11	$\overline{Q6}$	12	31	11				

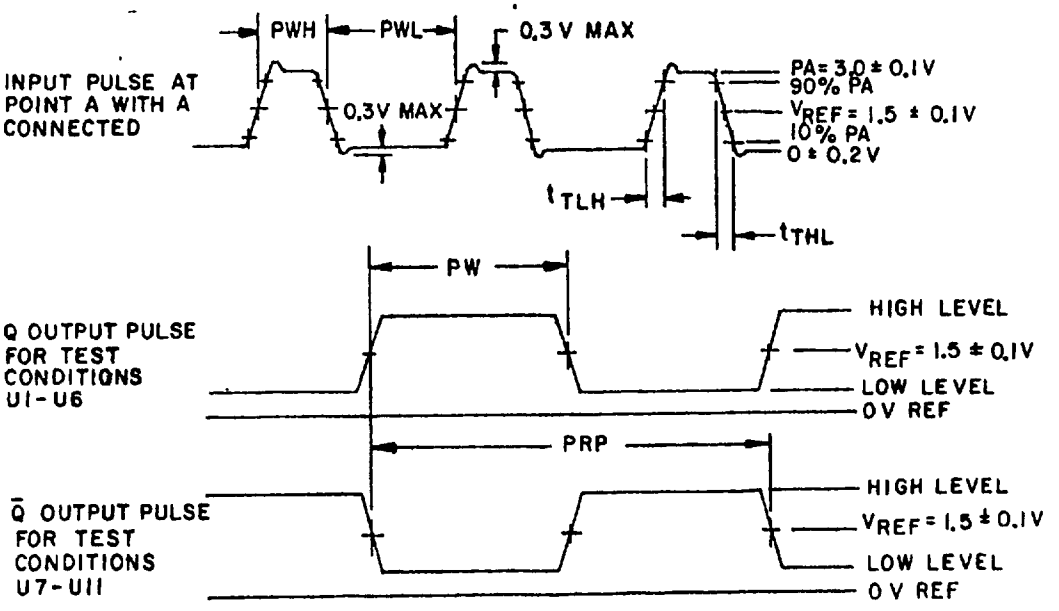
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NOTE R1 = 51.1Ω ±1%, 1 4 W.

FIGURE 20. Maximum operating frequency and minimum clock pulse width check test circuit for table XV.

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NOTES:
1. This test verifies minimum clock pulse width high level (12 ns), minimum clock pulse width low level (18 ns), minimum input setup time high level (12 ns), and minimum input hold time high level (0 ns).
2. Input pulse parameters:
   PWH = 13 ±1 ns.
   PWL = 19 ±1 ns.
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns.
3. Verify that all outputs have a high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V before making measurement.
4. All voltages are referenced to 0 V REF.

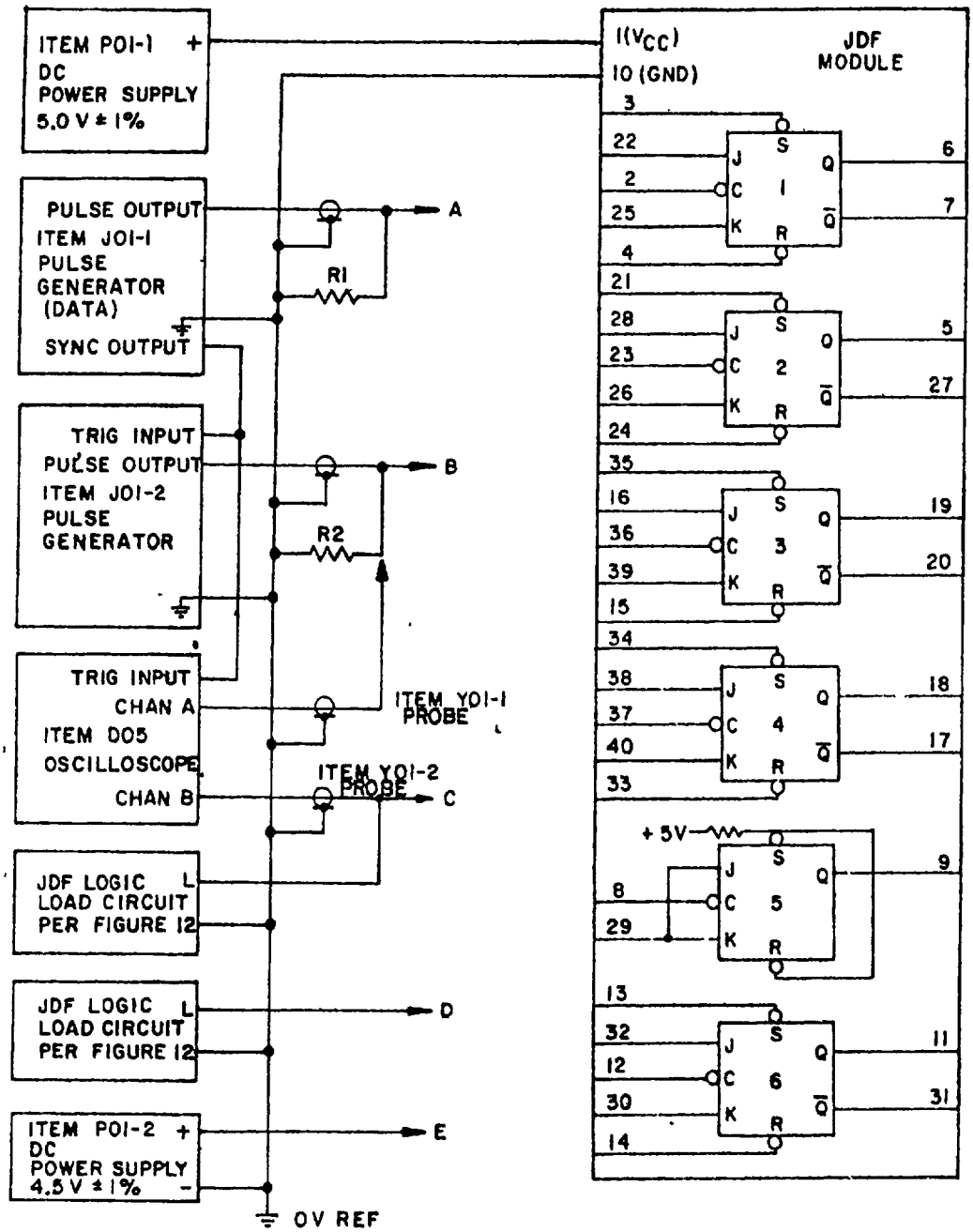
FIGURE 21. Maximum operating frequency and minimum clock pulse width check test circuit waveforms for table XV.

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TABLE XVI. Input low-level setup and hold time check

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	Q1 Input low-level setup and hold time (V_{OL} measurement)	22,25	2	6	7	3,4	0.0	0.6	V	22, 23
V2	Q2	28,26	23	5	27	21,24	↓	↓	↓	↓
V3	Q3	16,39	36	19	20	35,15				
V4	Q4	38,40	37	18	17	34,33				
V5	Q5	29	8	9	Open	Open				
V6	Q6	32,30	12	11	31	13,14				

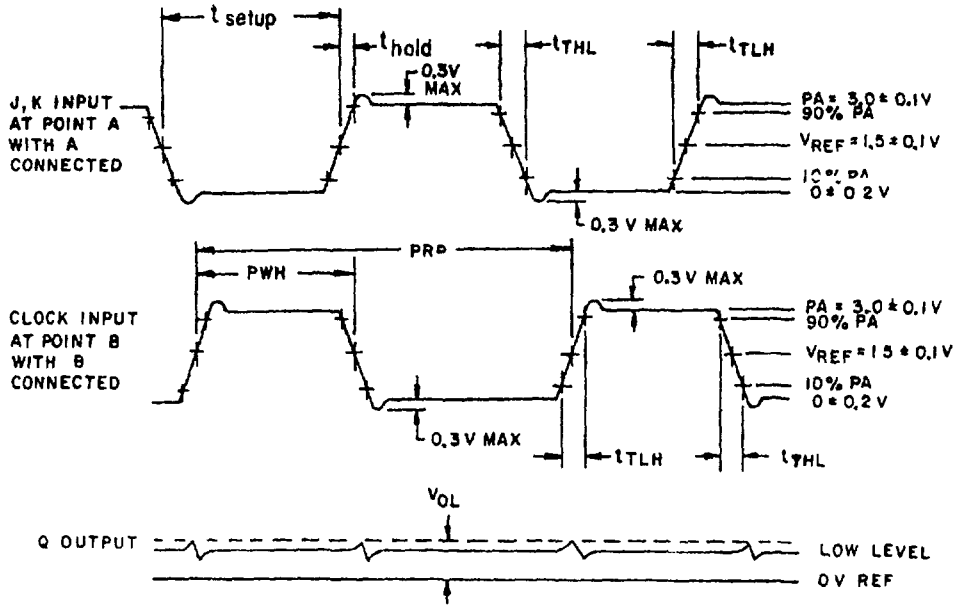
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NOTE: R1 = R2 = 51 1Ω ±1%, 1/4 W.

FIGURE 22. Input low-level setup and hold time check test circuit for table XVI.

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NOTES:
1. This test verifies an input low-level setup time of 15 ns and hold time of 0 ns.
2. Input pulse parameters:
   (a) PWH = 13 ±1 ns.
   (b) PRP = 50 ±5 ns.
   (c) t_{setup} = 16 ±1 ns.
   (d) t_{hold} = 1 ±1 ns.
   (e) $t_{TLH} = t_{THL}$ = 6 ±2 ns.
3. Initially set each circuit in a low state.
4. Using item D05 verify that the output (V_{OL}) is $\leq$ 0.6 V.

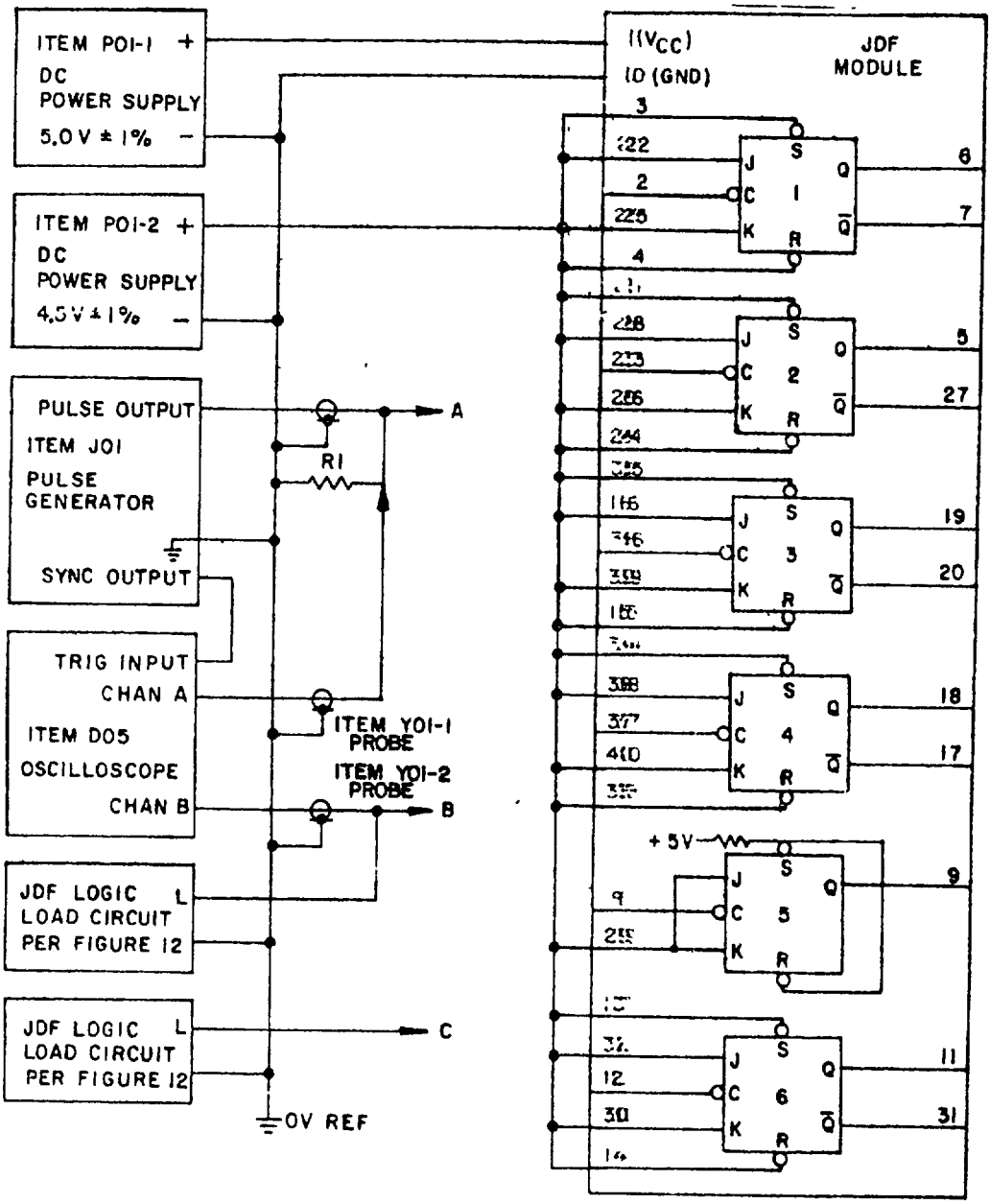
FIGURE 23. Input low-level setup and hold time check test circuit waveforms for table XVI.

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TABLE XVII. Maximum transition time check.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	Q1 Maximum transition time (PW measurement)	2	6	7	960	1,040	ns	24, 25
W2	Q2	23	5	27				
W3	Q3	36	19	20				
W4	Q4	37	18	17				
W5	Q5	8	9	Open				
W6	Q6	12	11	31				

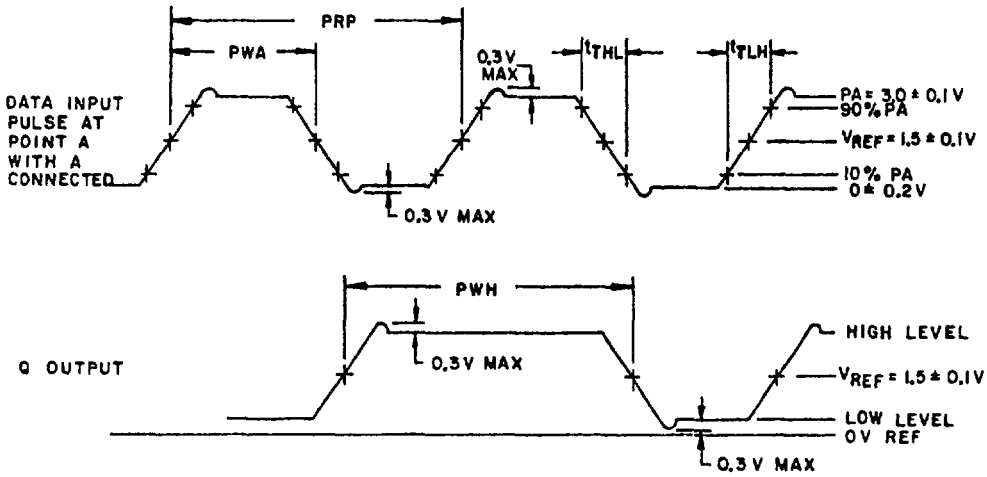
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 24. Maximum transition time check test circuit for table XVII.

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NOTES:
1. This test verifies toggle operation with a transition time of 120 ±5 ns.
2. Input pulse parameters:
   (a) PWA = 500 ±10 ns.
   (b) PRP = 1,000 ±100 ns.
   (c) $t_{TLH} = t_{THL} = 120 \pm 5$ ns.
3. Measure the output pulse width (PWH).
4. Verify that all outputs have high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V before making measurement.
5. All voltages are referenced to ground.

FIGURE 25. Maximum transition time check test circuit waveforms for table XVII.

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TABLE XVIII. Output crosstalk (low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X1	Q1 Output crosstalk, low level	3,21,24,35,15,34,33,29,13,14	4	6	0	680	mV	26, 27
X2	Q2	3,4,21,35,15,34,33,29,13,14	24	5				
X3	Q3	3,4,21,24,35,34,33,29,13,14	15	19				
X4	Q4	3,4,21,24,35,15,34,29,13,14	33	18				
X5	Q6	3,4,21,24,35,15,34,33,29,13	14	11				
X6	[illegible]	4,21,24,35,15,34,33,29,13,14	3	7				
X7	Q2	3,4,24,35,15,34,33,29,13,14	21	27				
X8	Q3	3,4,21,24,15,34,33,29,13,14	35	20				
X9	Q4	3,4,21,24,35,15,33,29,13,14	34	17				
X10	Q6	3,4,21,24,35,15,34,33,29,14	13	31				
X11	Q1	3,21,35,15,34,33,29,13,14	4,24	6		360		
X12	Q2	3,21,35,15,34,33,29,13,14	4,24	5				
X13	$\bar{Q}1$	4,24,35,15,34,33,29,13,14	3,21	7				
X14	$\bar{Q}2$	4,24,35,15,34,33,29,13,14	3,21	27				
X15	Q3	3,4,21,24,35,34,29,13,14	15,33	19		320		
X16	Q4	3,4,21,24,35,34,29,13,14	15,33	18				
X17	$\bar{Q}3$	3,4,21,24,15,33,29,13,14	35,34	20				
X18	$\bar{Q}4$	3,4,21,24,15,33,29,13,14	35,34	17				
X19	Q6	3,4,21,24,35,15,34,33,13	29,14	11		310		
X20	$\bar{Q}6$	3,4,21,24,35,15,34,33,14	29,13	31		310		

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TABLE XIX. Output crosstalk (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y1	Q1 Output crosstalk, high level	4,21,24,35,15,34,33,29, 13,14	3	6	0	400	mV	26, 27
Y2	Q2	3,4,24,35,15,34,33,29,13, 14	21	5				
Y3	Q3	3,4,21,24,15,34,33,29,13, 14	35	19				
Y4	Q4	3,4,21,24,35,15,33,29,13, 14	34	18				
Y5	Q6	3,4,21,24,35,15,34,33,29, 14	13	11				
Y6	$\overline{Q1}$	3,21,24,35,15,34,33,29, 13,14	4	7				
Y7	$\overline{Q2}$	3,4,21,35,15,34,33,29,13, 14	24	27				
Y8	$\overline{Q3}$	3,4,21,24,35,34,33,29,13, 14	15	20				
Y9	$\overline{Q4}$	3,4,21,24,35,15,34,29,13, 14	33	17				
Y10	$\overline{Q6}$	3,4,21,24,35,15,34,33,29, 13	14	31				

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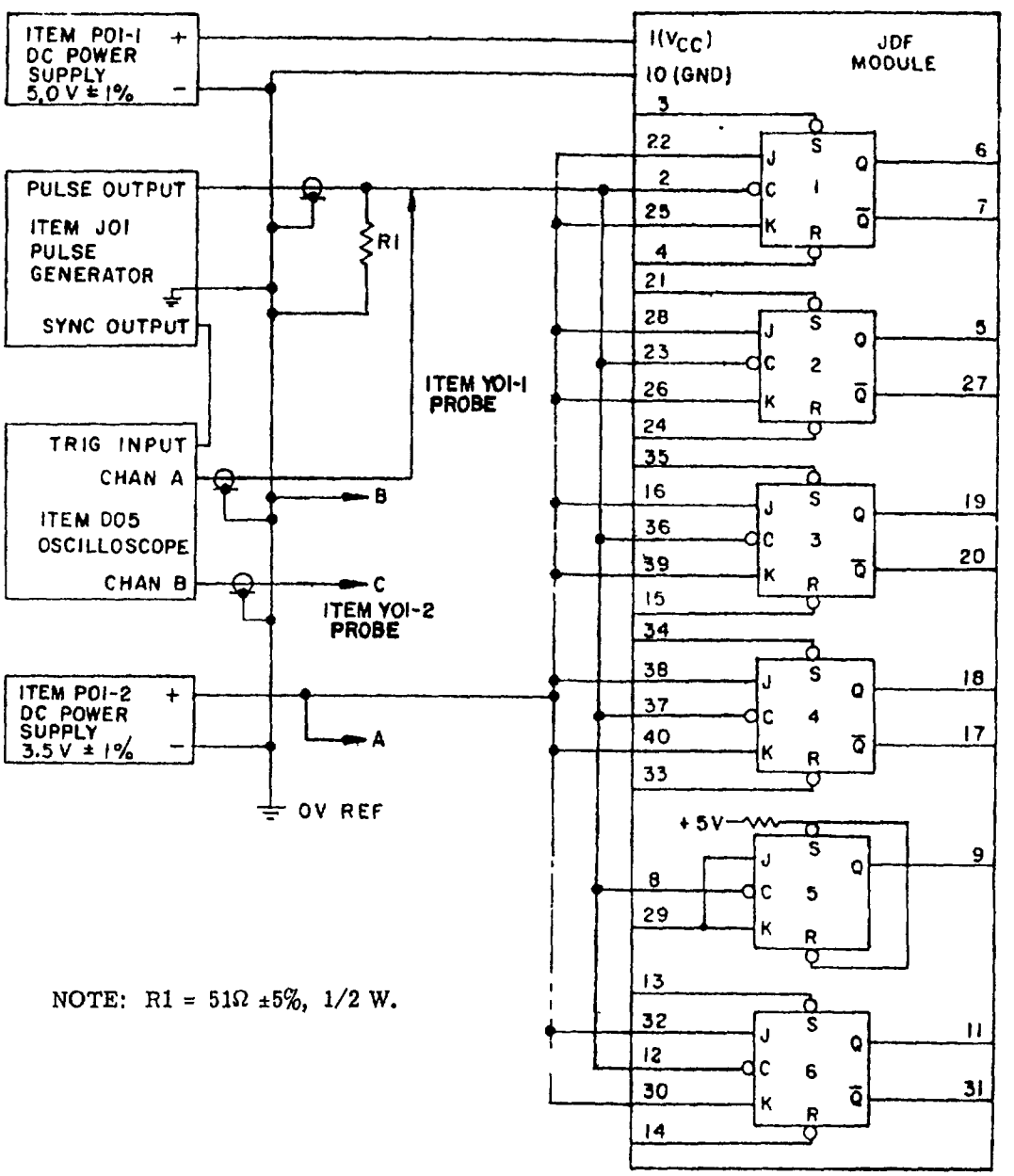
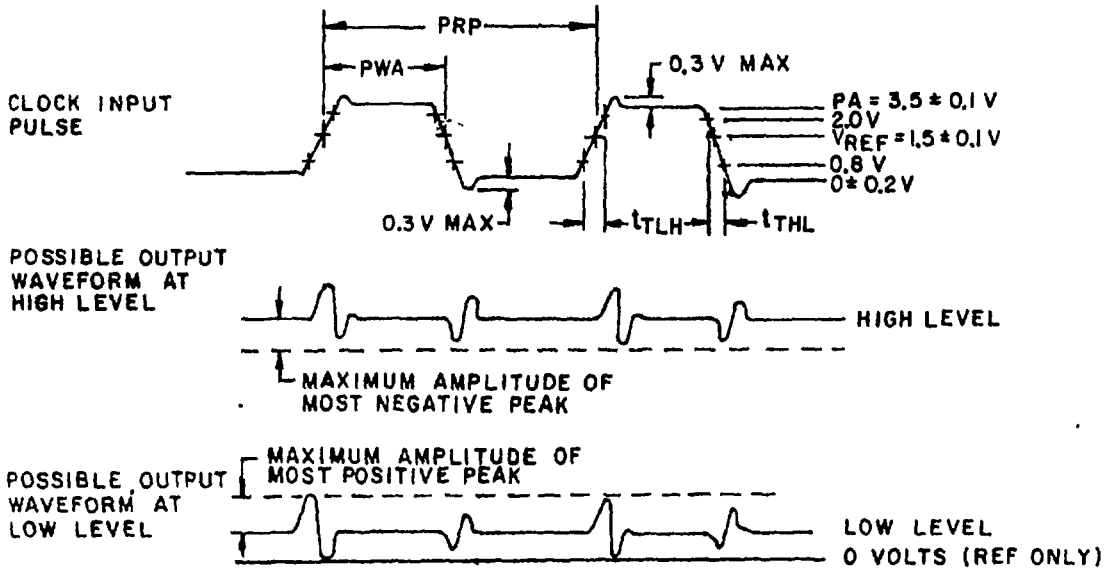


FIGURE 26. Output crosstalk (low-level and high-level) test circuit for tables XVIII and XIX.

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NOTE: Clock input pulse parameters:
PRP = 500 ±100 ns.
PWA = 250 ±50 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at all input pins).

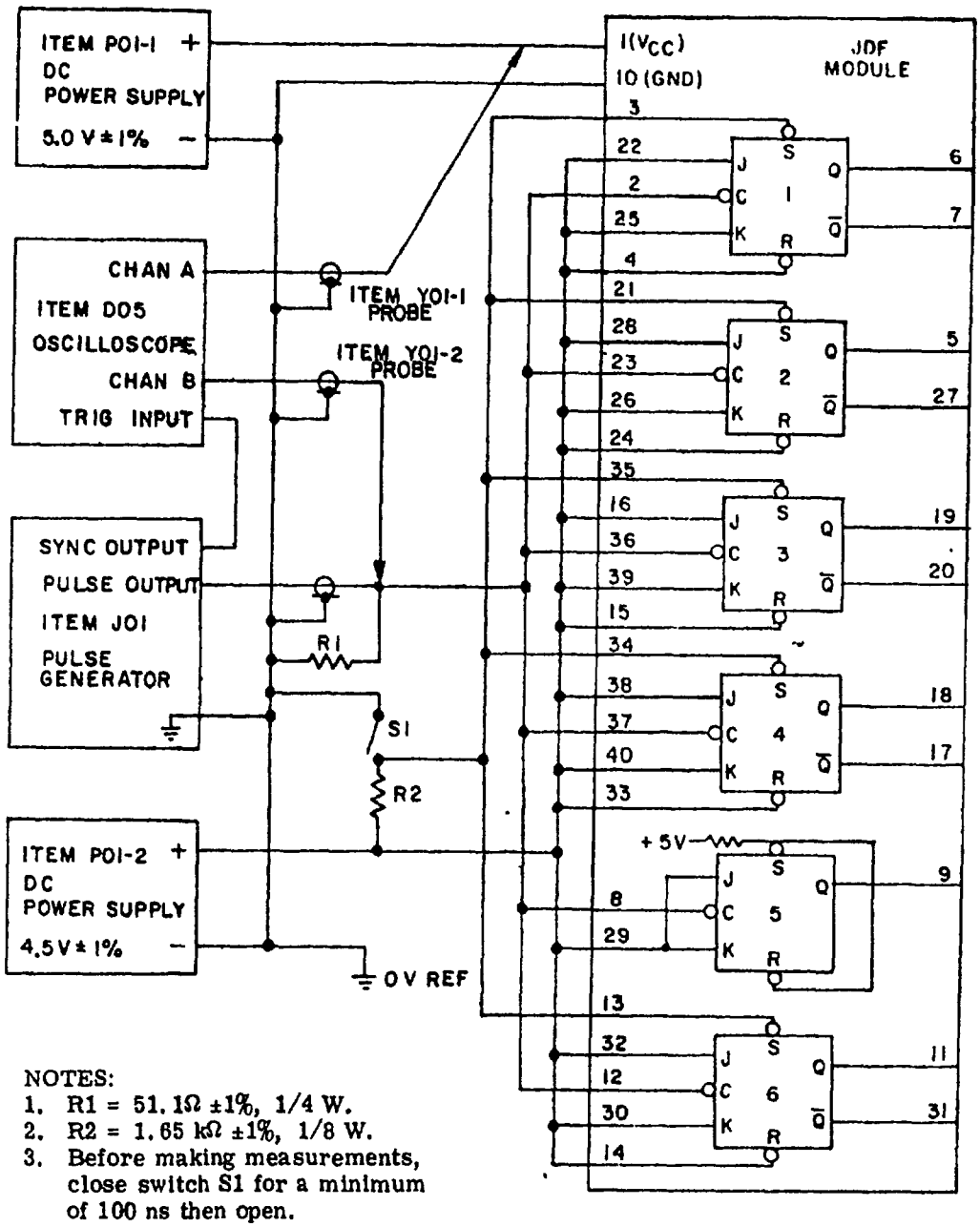
FIGURE 27. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVIII and XIX.

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TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
Z	Power supply transient amplitude	0	400	mV	28, 29

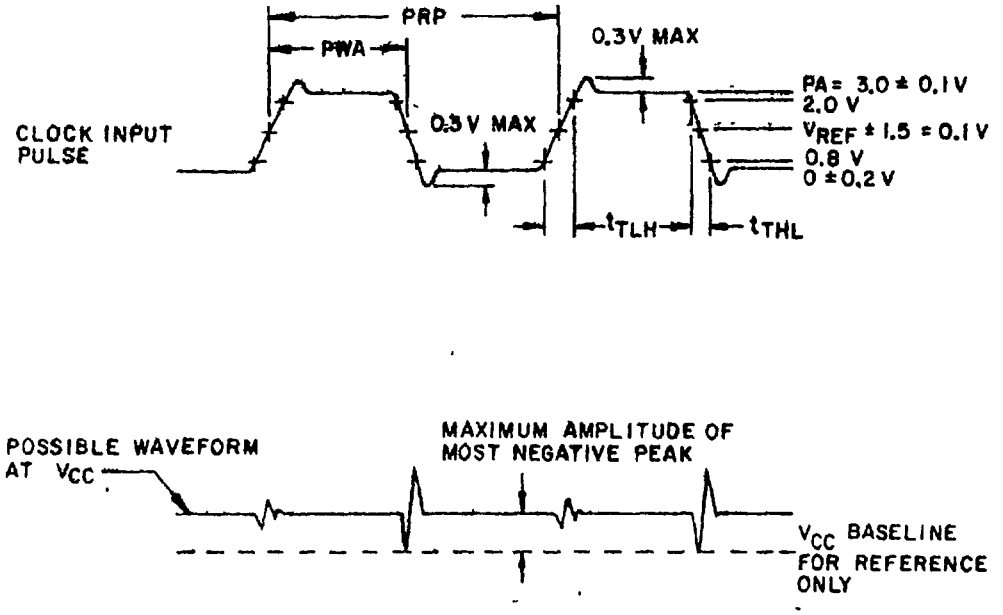
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. R2 = 1.65 kΩ ±1%, 1/8 W.
3. Before making measurements, close switch S1 for a minimum of 100 ns then open.

FIGURE 28. Power supply transient amplitude test circuit for table XX.

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NOTES:
1. Clock input pulse parameters:
   (a) PRP = 500 ±100 ns.
   (b) PWA = 250 ±50 ns.
   (c) $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at all input pins).
2. All voltages are referenced to ground.

FIGURE 29. Power supply transient amplitude test circuit waveforms for table XX.

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TABLE XXI. Power supply filter capacitance test.

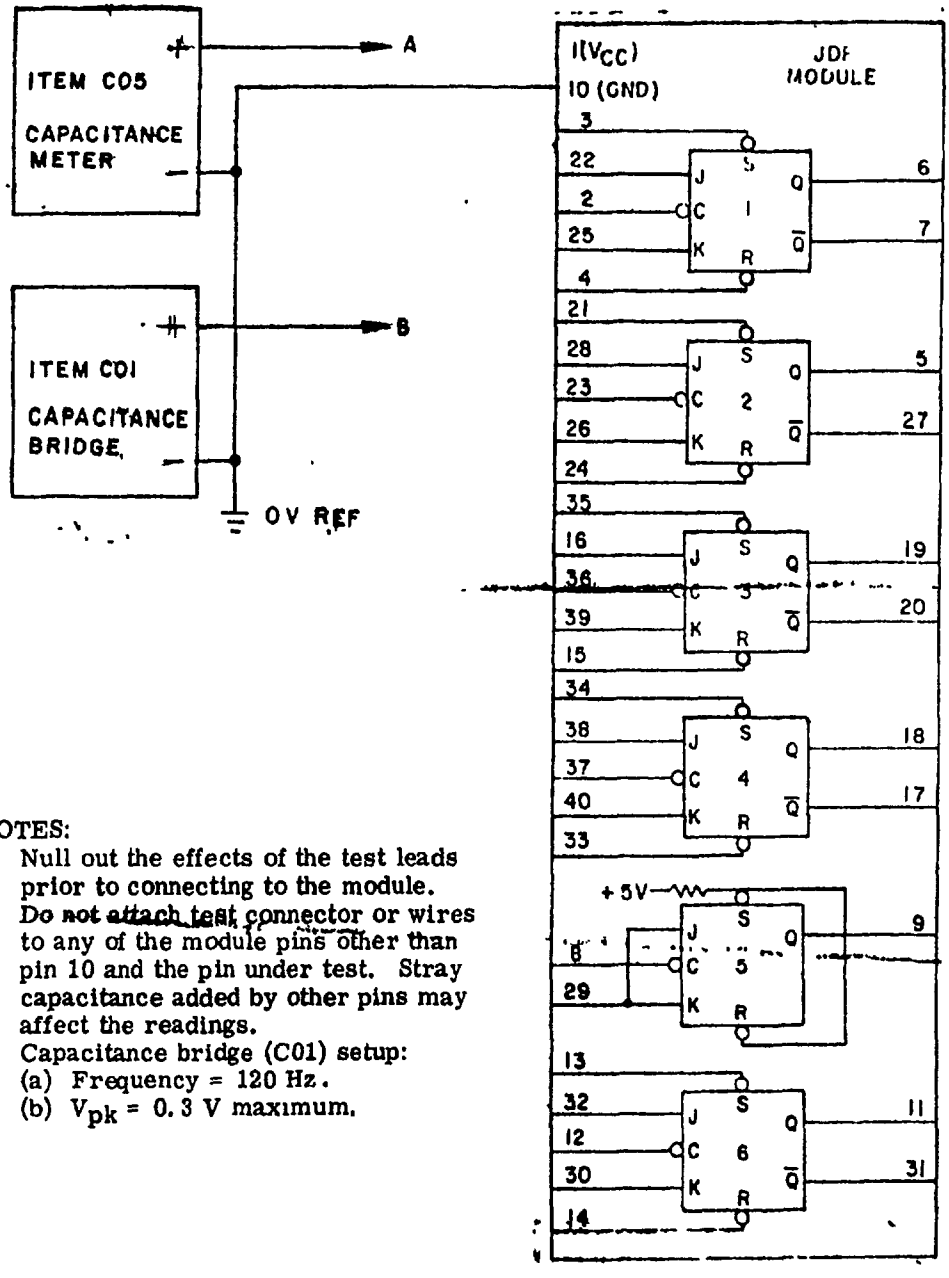
Test condition	Parameter	Connect		Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA	Power supply filter capacitance	Open	1	1.6	3.0	µF	30

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TABLE XXII. Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
BB1	J1	Input capacitance	22	Open	2	15	pF	30
BB2	J2		28					
BB3	J3		16					
BB4	J4		38					
BB5	J6		32					
BB6	K1		25					
BB7	K2		26					
BB8	K3		39					
BB9	K4		40					
BB10	K6		30					
BB11	R1		4					
BB12	R2		24					
BB13	R3		15					
BB14	R4		33					
BB15	R6		14					
BB16	S1		3					
BB17	S2		1					
BB18	S3		35					
BB19	S4		34					
BB20	S6		13					
BB21	C1		2					
BB22	C2		23					
BB23	C3		36					
BB24	C4		37					
BB25	C5		8					
BB26	C6		12					
BB27	JK5		29		4	30		

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NOTES:
1. Null out the effects of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added by other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   (a) Frequency = 120 Hz.
   (b) V_{pk} = 0.3 V maximum.

FIGURE 30. Power supply filter and input capacitance test circuit for tables XXI and XXII.

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